

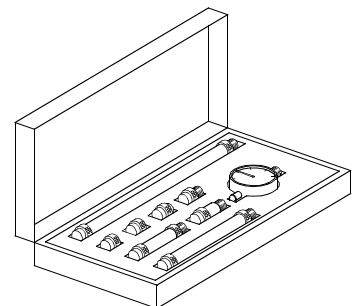
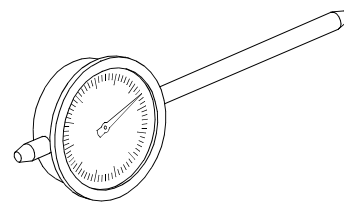
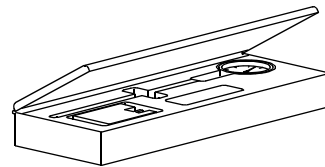
SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
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Standard Tools: See Section 913





Spare Parts

Plate – Item No.	Description	Qty
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General

The condition of the main bearings can be checked by:

- Deflection readings
- Visual checking
- Edge checking and
- Measuring of top clearance.

Crankshaft deflection readings

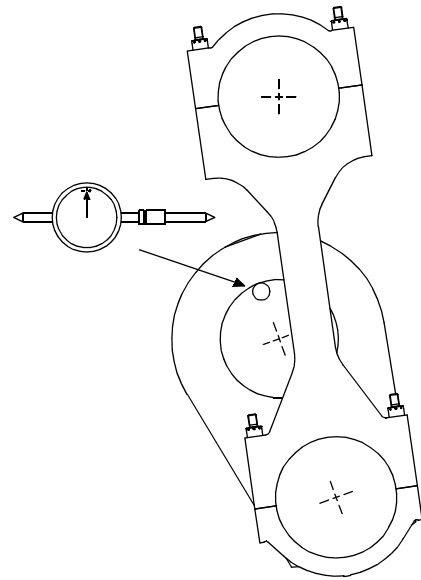
Crankshaft deflection readings should be taken while the ship is afloat (i.e. not while in dry dock).

As the alignment is influenced by the engine temperature as well as the loading conditions, deflection measurements should, for comparison purposes, always be made under nearly the same temperature and load conditions.

1. Place a dial gauge axially in the crank throw opposite the crankpin, as illustrated on the sketch. The correct mounting position is marked with punch marks on the crank throw. See also Chapter 708 in Volume I.
2. "Closing" of the crank throw (compression of the gauge) is regarded as negative (–) and "opening" as positive (+).
3. Set the dial gauge to zero at the **B1** side near BDC. Whilst turning clockwise, take the readings when the throw passes the positions:

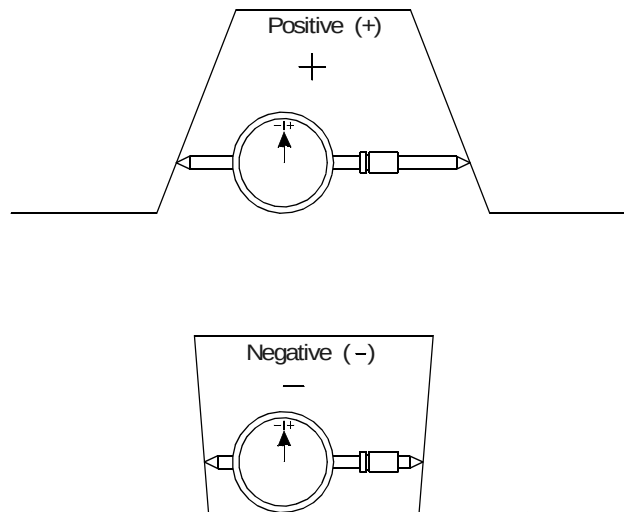
- B1** (near bottom)
C (camshaft side)
T (top)
E (exhaust side)
B2 (near bottom)
 $1/2 (B2+B1) = B$

1.



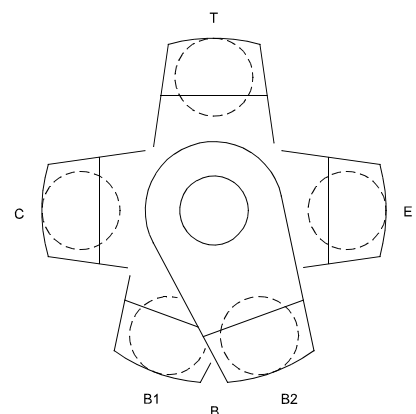
HM905-2.0 46 01

2.



KN905-1.1 208 02

3.

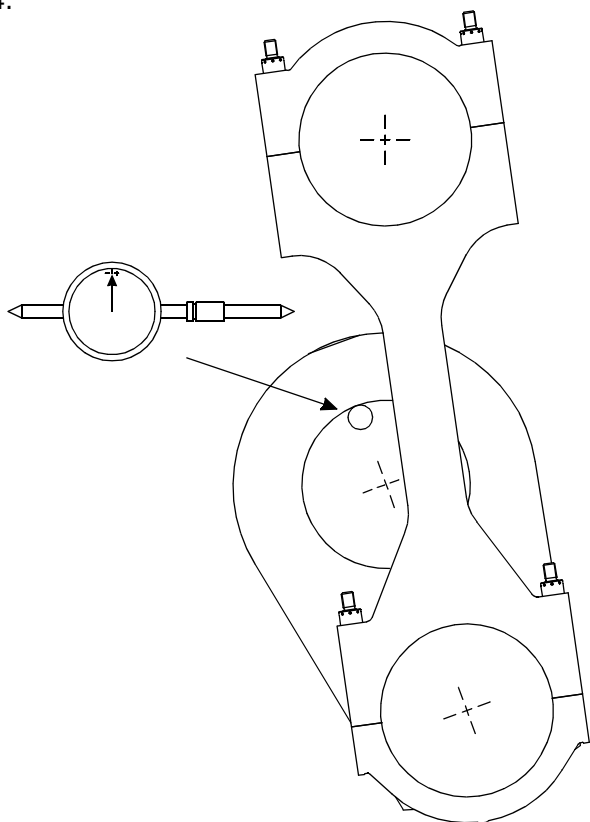


Looking forward

B1
C
T
E
B2
$1/2 (B1 + B2) = B$

XZ905-1.1 204 03

4.



HM905-2.0 46 01

4. When taking deflection readings for the three aftmost cylinders, the turning gear should, at each stoppage, be turned a little backwards to ease off the tangential pressure on the turning wheel teeth. Otherwise, this pressure may falsify the readings.

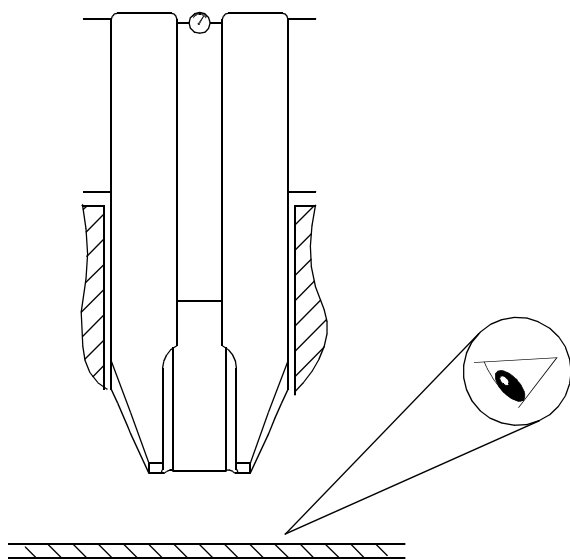
For evaluation of the crankshaft deflection readings, see *Volume I, OPERATION, Chapter 708*.

If the crankshaft deflection (alignment indicator) is approaching the tolerance limits (see *Vol. I, OPERATION*), the two adjacent main bearings must be checked for wear. See next page for checking of main bearing.

If the bearings are found to be in good order, please contact MAN B&W Diesel or the engine manufacturer for checking of the bedplate alignment.

As reference, use the "after seatrial" deflection table.

5.



KN905-1.1 208 05

The condition of the bearing can be checked as follows:

5. Visually look for bearing metal fragments in the oil pan, and check the filters for metal fragments.

Such fragments can be from crosshead, guide shoe or crankpin bearings. If fragments are found, the damaged bearing can be found by edge checking with a feeler.

General

Bearing damage usually propagates rather quickly towards the edge of the bearing where, eventually, it causes chips to be broken off, which means that loose pieces of white metal can fall down into the crankcase beneath the bearing support.

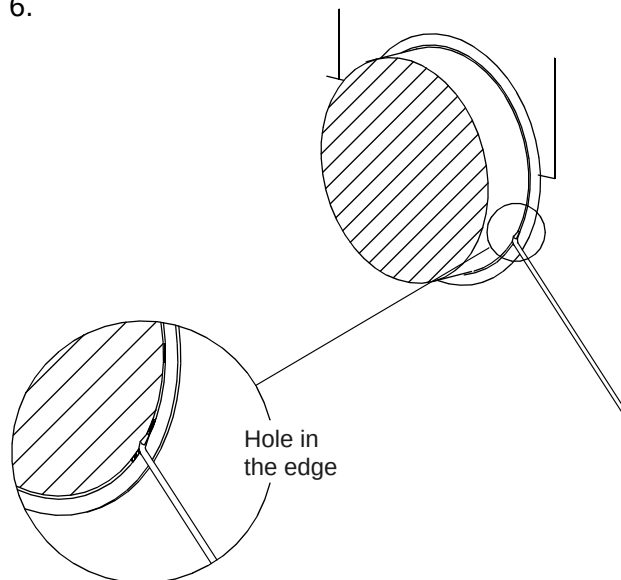
Large, thick pieces will normally be found during a crankcase inspection, and small, thin pieces can enter the filter. A check of the bearing edges together with inspection of the crankcase and the filter normally provides a good indication of the bearing condition.

Checking the bearing edges

6. Bearing edges can be checked with a “feeler” that is able to follow the bearing edge against the journal the whole way round on each side. If white metal is missing at the edge, the tip of the “feeler” will enter the hole, thus locating the damage. In most cases this hole can be seen by the naked eye as a dark spot when using a strong flashlight.

The No. 1 and the two aftmost main bearings are more difficult to access on the whole circumference on each side, but it is often possible to bend the “feeler” to suit the situation and thus reach as far round as possible.

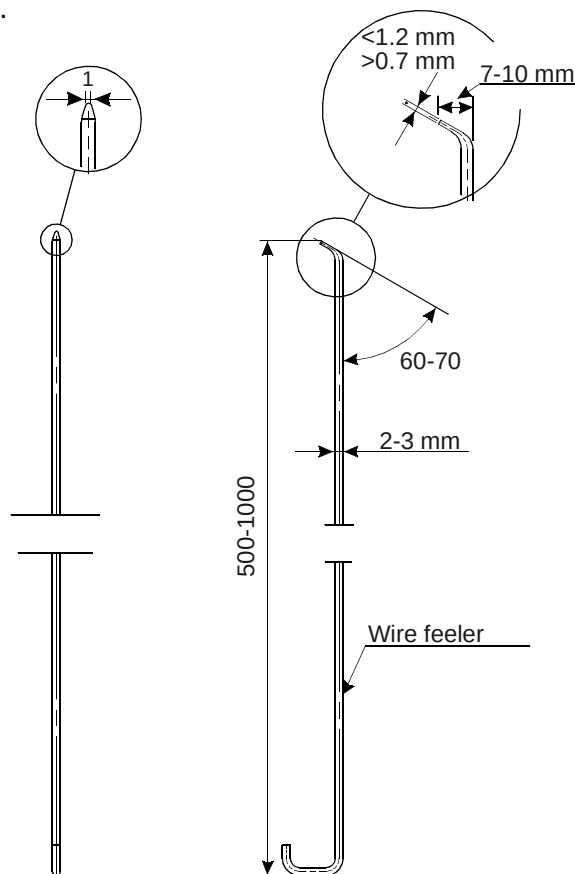
6.



Wire-feeler(made by crew)

7. It is rather easy to make a feeler (see sketch). The “feeler” should be made from a piece of steel wire (e.g. a welding rod) approx. 0.6 - 1 m long depending on the engine type, and 2 to 3 mm thick. Approx. 7 to 10 mm of the wire should be bent to an angle of approx. 65° to form a “feeler tip”. Grind the tip smooth to obtain the shape and dimension shown on the sketch. Note that the thickness of the white metal is approx. 1.5 mm, which is why the tip should be less than 1.2 mm thick, and bigger than the max. top clearance. At the other end of the tip, a “handle” should be made by bending a hook or similar in the same direction as the tip.

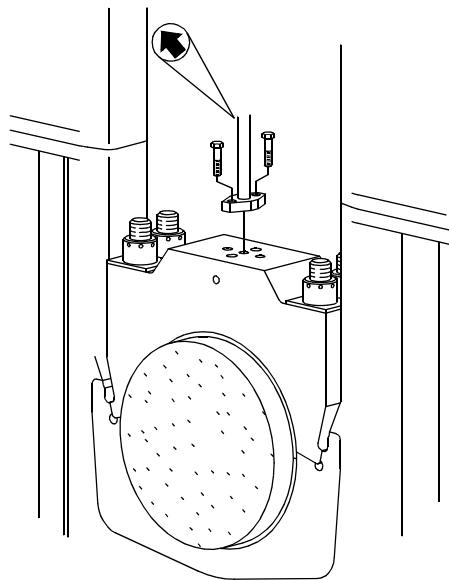
7.



Note!

The above dimensions of the “wire-feeler” are guideline values and may depend on the engine type as well as individual, personal designs.

8.



8. Remove the lub. oil pipe from the main bearing cap.

Calibrate the main bearing measuring tool:

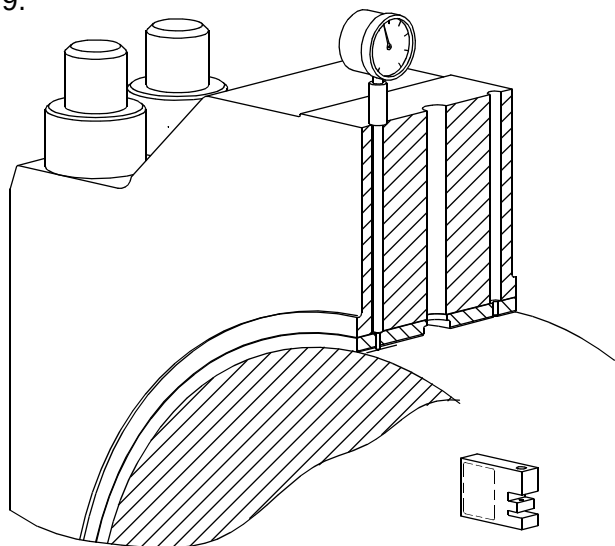
See the wooden box for the measuring tool.

9. Insert the measuring tool and check the clearance between the upper shell and the journal.

Note!

The bearing clearances for a new bearing stated in Data (see *Data 105-2*) are for guidance purposes only.

9.

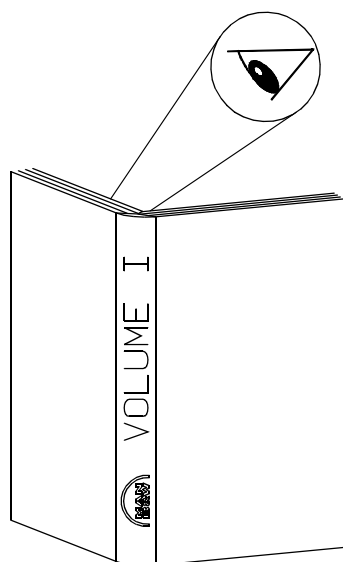


10. The difference between the actual clearance measurement and the measurement recorded in the adjustment sheet (or the clearance noted for a new bearing installed later) must not exceed 0.10 mm. If it does, see *Volume I, Operation, Chapter 708*.

For evaluation of the bearing, see *Volume I, Operation, Chapter 708*.

If the bearing needs to be disassembled for inspection, see procedure 905-2.2.

10.



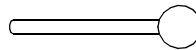
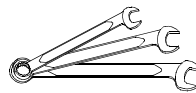
SAFETY PRECAUTIONS

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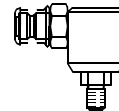
Data

Ref.	Description	Value	Unit
	Top clearance in main bearing		
	Max.	0.65	mm
	Min.	0.35	mm
D-1	Main bearing cap	290	kg
D-2	Main bearing shells		
	– upper	20	kg
	– lower	20	kg
D-3	Hydraulic pressure		
	– dismantling	1500-1650	bar
	– tightening	1500	bar

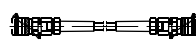
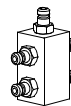
Standard Tools: See Section 913



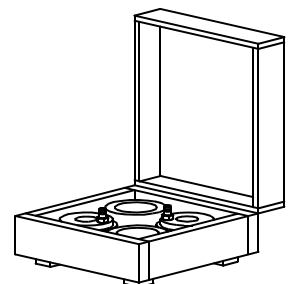
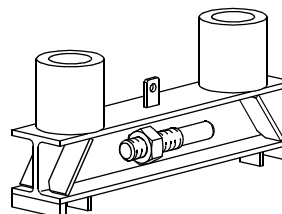
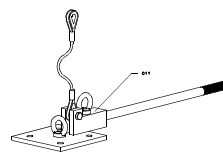
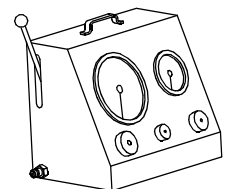
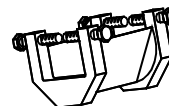
2 x 600 kg

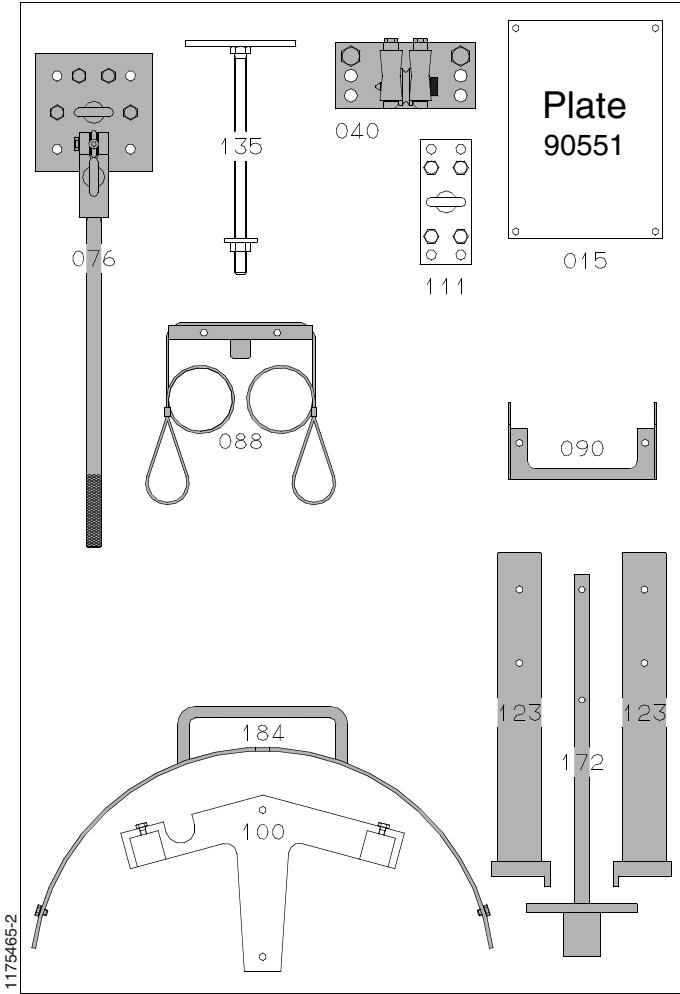


x 2


4 x 1,5 m
1 x 3 m


2 x 500 kg





Spare Parts

Plate – Item No.	Description	Qty
91210-xxxx 050	Screw	2
91210-xxxx 108	Main bearing, upper shell	1
91210-xxxx 121	Main bearing, lower shell	1

Positioning the crankshaft

1. Dismantling of main bearing.

The bearings are divided into two categories:

- The main bearings
- The aftmost main bearing often referred to as "Journal bearing" (See Procedure 905-4).

2. Before any dismantling, check and write down the main bearing top clearances and crankshaft deflection readings for the cylinder unit concerned.

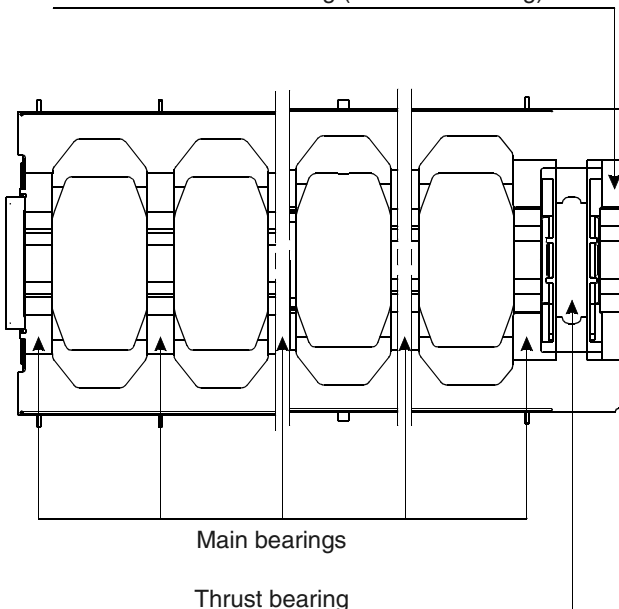
3. Turn the crank throw to the position shown on the sketch and in such a way that the top of the studs is flush with the crank throw.

Note!

Be sure that the crosshead for the neighbouring cylinder is in a higher position than for the cylinder you are working on, otherwise it will be impossible to lift the cap.

1.

Aftmost main bearing (= Journal bearing)



KN905-2.2 238 01

2.

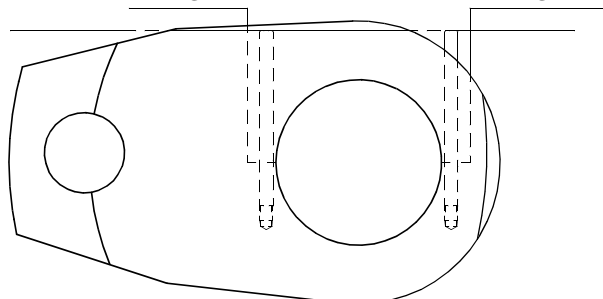
AUTOLOG + CLEARANCES									
CYL 1									
CYL 2									
CYL 3	0,0	0,0	0,0	0,0	0,0				
CYL 4									
CYL 5									
CYL 6									
CYL 7									
CYL 8									

KN905-2.2 238 02

3.

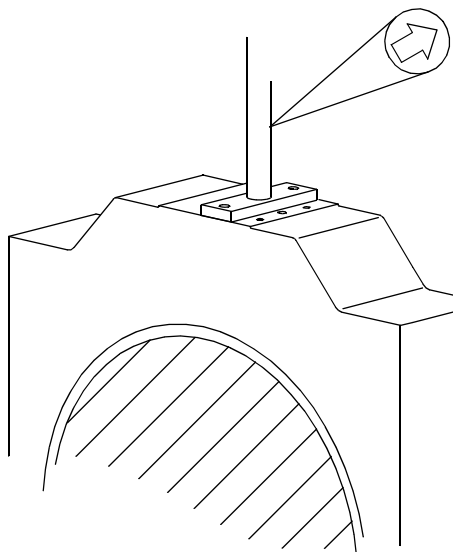
EXH-SIDE

MAN-SIDE



KN905-2.2 238 03

4.



KN905-2.2 238 04

Oil pipes

4. Disconnect the lubricating oil pipe from the main pipe.

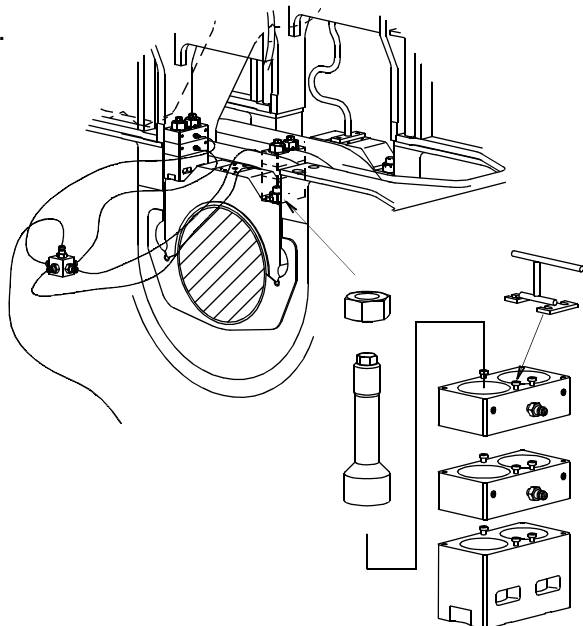
Note!

On some engines, the oil pipes for the Axial Vibration Damper (AVD) have to be removed together with the oil pipe on main bearing No. 1.

Note!

Some engines have temperature sensors fitted to the main, crosshead, and crank-pin bearings. These sensors must be carefully handled to avoid damaging the equipment.

2.



GN905-2.2 204 02

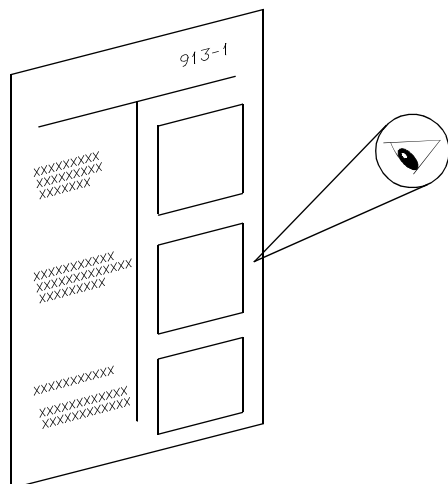
5. Mount the extension studs, the spacers, the hydraulic jacks and the nuts on the extension studs. Loosen the main bearing stud nuts, using the hydraulic jacks.
For operation of the hydraulic jacks, see Procedure 913-1.

6. Remove the hydraulic main bearing tools from the engine.

Unscrew and remove the nuts from the main bearing studs.

Remove the rectangular nut washers from the bearing cap.

5.



KN905-2.2 238 07

7. Mount the lifting tool on the main bearing cap.

Mount the pulley on the web-plate over the main bearing, and mount the wire rope from the lifting tool in the pulley.

Mount an eye bolt in one of the threaded holes on the opposite web-plate.

8. Mount the wire guide as shown.

Move the lifting pipes below the platform outside the relevant cylinder.

Mount tackles in the following positions:

A + B: Two 2-ton chain tackles, below the platform.

C: One 1-ton chain tackle.

D: One ½-ton chain tackle, in the eye bolt on the opposite web-plate.

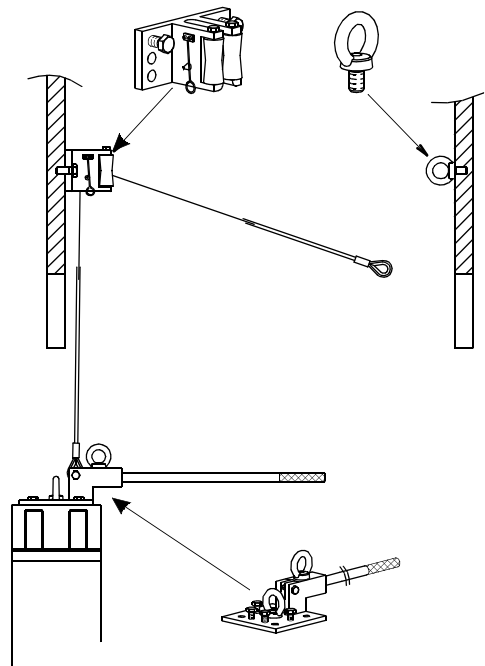
9. Attach the wire rope from the lifting tool to tackle **A**.

Using tackle **A**, lift the main bearing cap free of the studs.

Note!

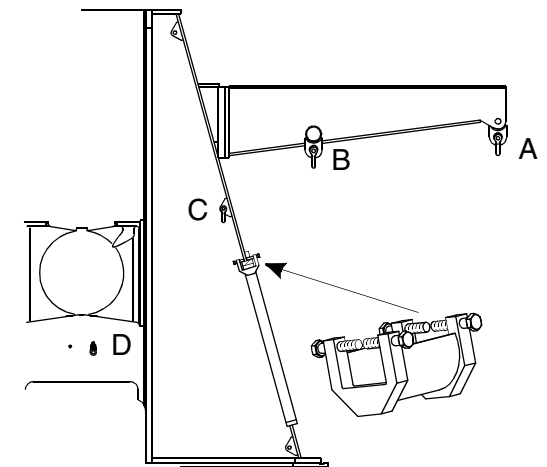
Owing to space requirements below the web-plate, it might be necessary to dismount the eye bolt from the lifting tool.

7.



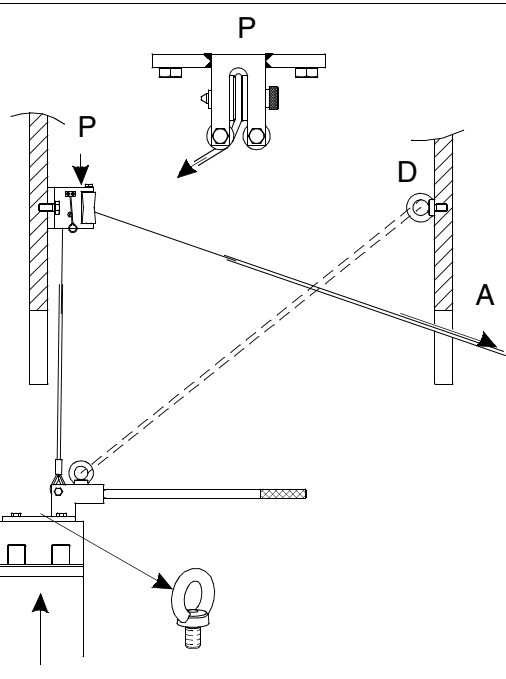
GN905-2.2 241 07

8.

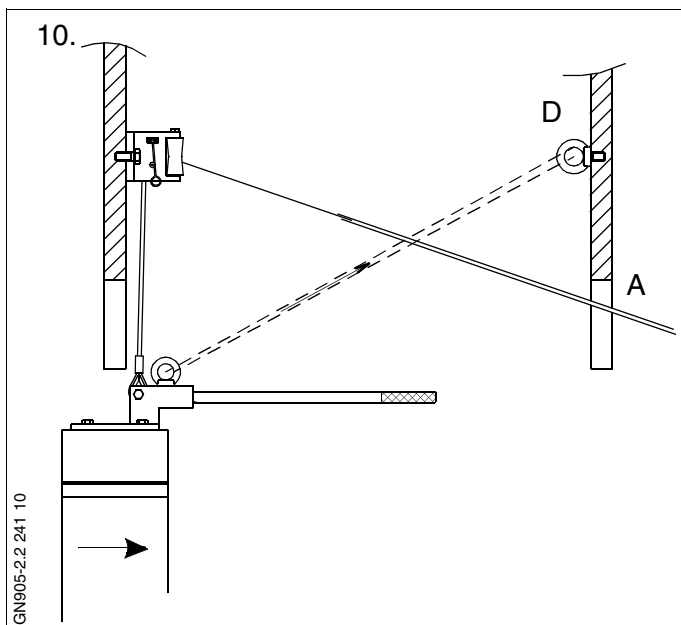


GN905-2.2 241 08

9.



GN905-2.2 241 09



10. Pull the bearing cap clear of the crosshead guides by using chain tackle **D**.

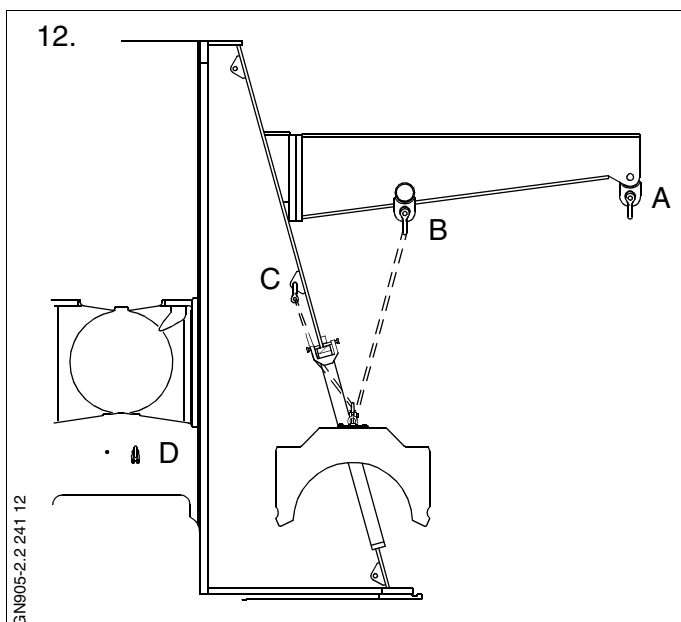
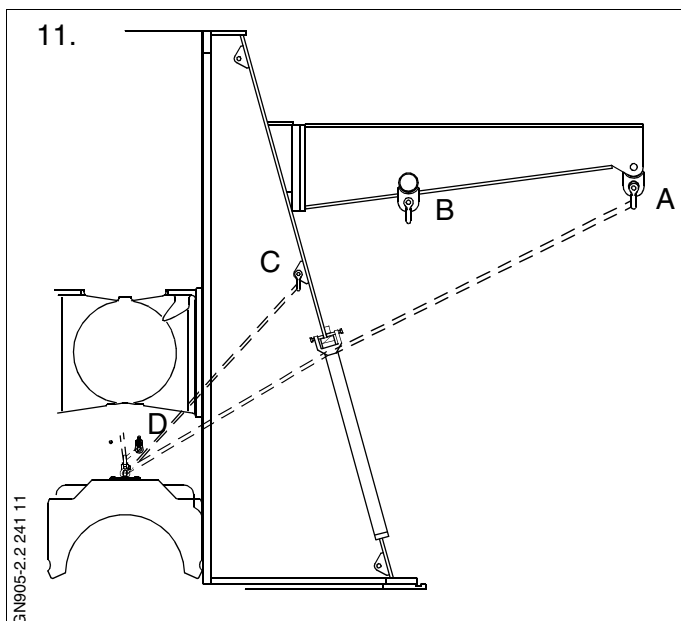
11. Attach tackle **C** to the eye bolt in the centre of the lifting tool.

Dismount the balancing handle from the lifting tool for main bearing cap.

Tighten tackle **C**, and release tackles **A** and **D**.

When released, dismount tackle **A** from the wire rope, and dismount the wire rope from the pulley.

12. Lift out the bearing cap, using tackles **B** and **A**.



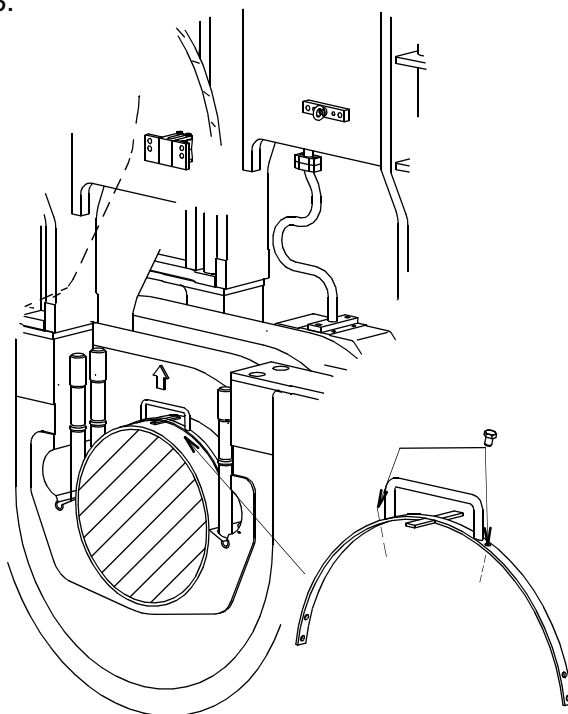
13. Mount the lifting tool on the upper main bearing shell, and lift out the shell by hand.
14. If the crankshaft is turned with the bearing cap dismounted, use the two stops to prevent the lower shell from being rolled out.
15. Place the crosspiece in the bedplate with the ends resting on the cross girders, and position the hydraulic jacks beneath the crank webs as shown.

Tighten the screws against the crank webs and compress the jacks.

Connect the hydraulic jacks to the high-pressure pump, and raise the pressure until the crankshaft has been lifted 0.2-0.5 mm, but max. the clearance in the two adjacent bearings.

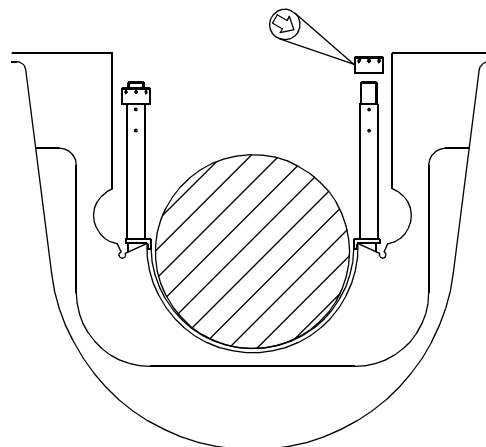
Note down the pressure for later reference.

13.



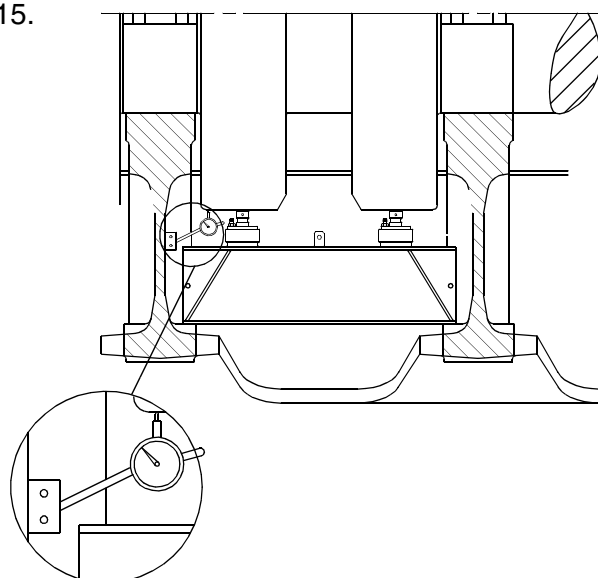
EN905-2.2 206 08

14.

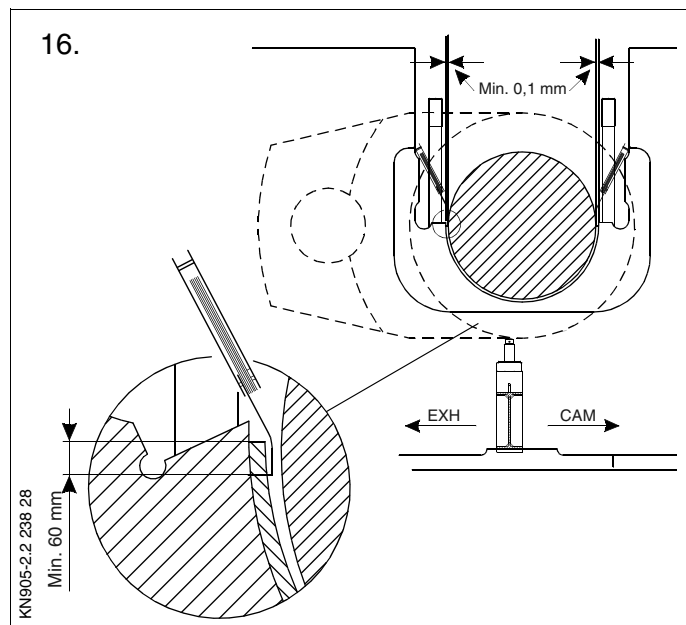


KN905-2.2 238 24

15.



KN905-2.2 238 27

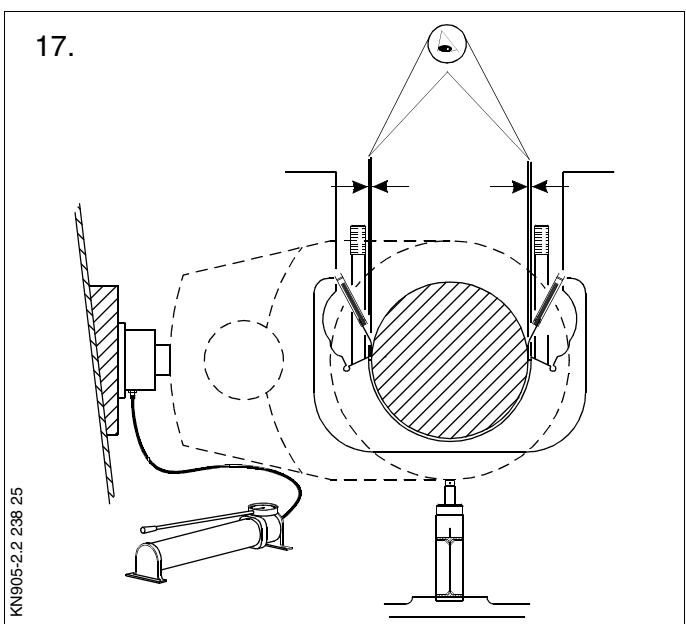


16. Check that there is min. 0.1 mm clearance between journal and bearing shell, or between bearing shell and main bearing support, on both sides of the journal.

Note!

The feeler blade must be inserted minimum 60 mm into the gap to pass the bore relief in the bearing shell.

If the clearance is less than 0.1 mm, move the crosspiece sideways until clearance is obtained. Normally, adjustment needs to be 50-150 mm to the exhaust side (the side where the clearance is missing).



Note!

The lower shell must be lifted out to the side with the most clearance, which is normally the camshaft side.

17. If re-positioning of the crosspiece does not ensure a vertical lift, place a 5-tonne jack between the side wall and the crank throw, to correct the journal position in this way.

If it is available, note down the pressure applied to the jack. This data is used to ensure that the crankshaft is in the same position when the new bearing is installed.

Note!

It is recommended that the crankshaft lifting tool is only removed if it is urgently necessary and then only after the main bearing shell has been reinstalled.

18. Dismount the locking screws.

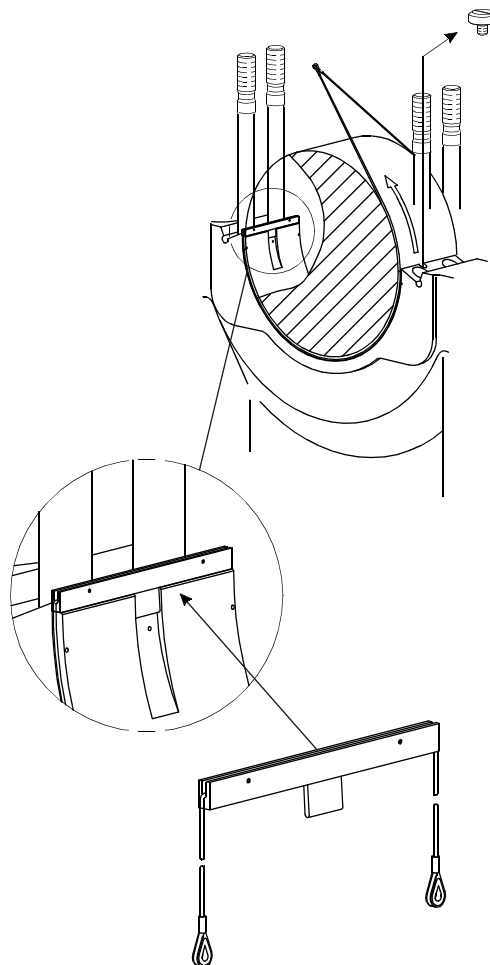
Place the dismantling tool on top of the lower bearing shell. Make sure that the flap on the dismantling tool enters the oil groove in the bearing shell.

19. Pull the lower shell around and up until it lies over the main bearing journal. Never use a tackle larger than 0.5 tonne.

Note!

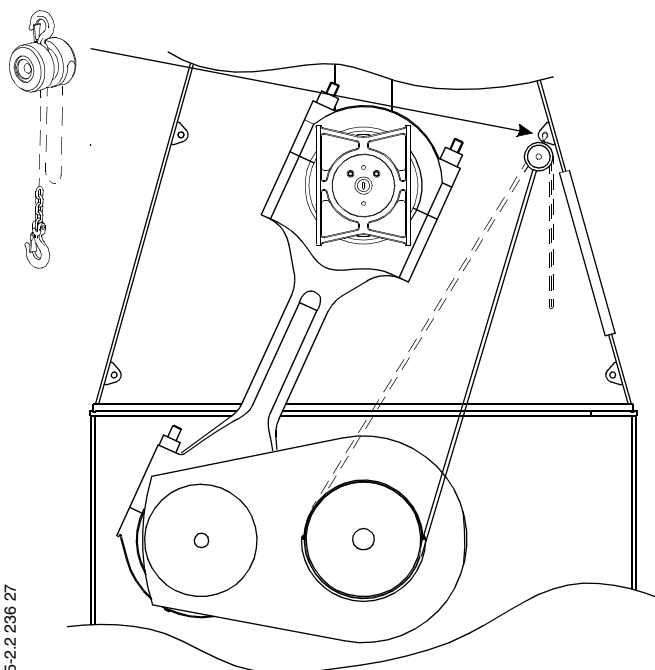
While pulling out the shell, it is recommended to hold a foot or hand on the wire rope, to tighten it up and to observe whether the shell is sticking.

18.



KN905-2.2 238 26

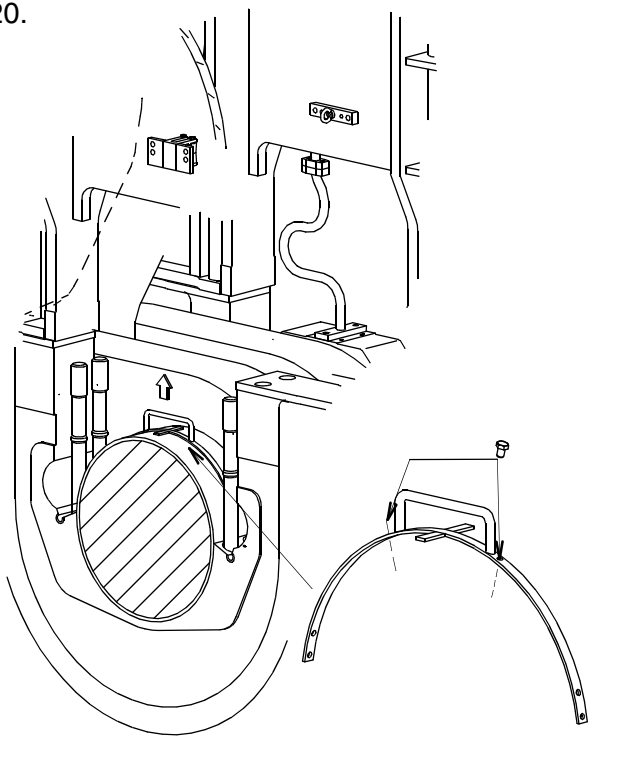
19.



JM905-2.2 236 27

20.

EN905-2.2 206 08



20. Take the lower shell out of the engine in the same way as the upper shell, i.e. using the lifting tool.

1. Clean and inspect the bearing shells. It is recommended that the main bearing shells be replaced in pairs.

If it is nevertheless desired to replace only one shell, this requires a careful evaluation of the condition of the shell that is to be re-installed.

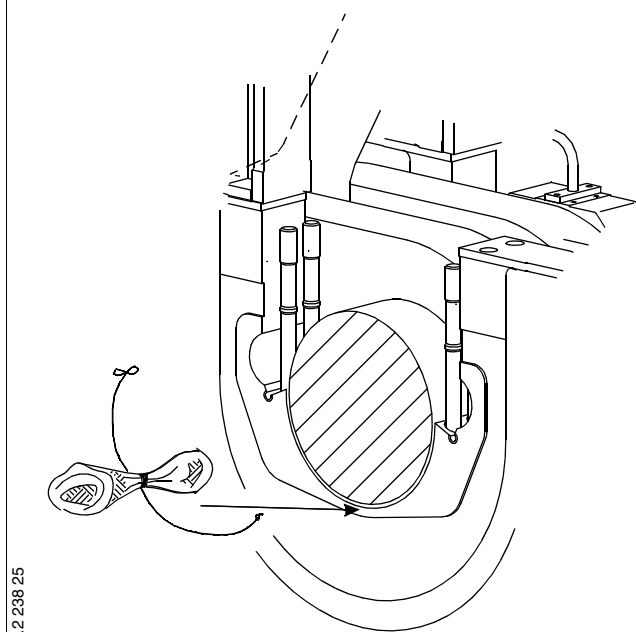
See Instruction book, Volume I, Chapter 708.

For advice on replacing individual bearing shells, it is recommended to contact MAN B&W Diesel A/S or the engine builder.

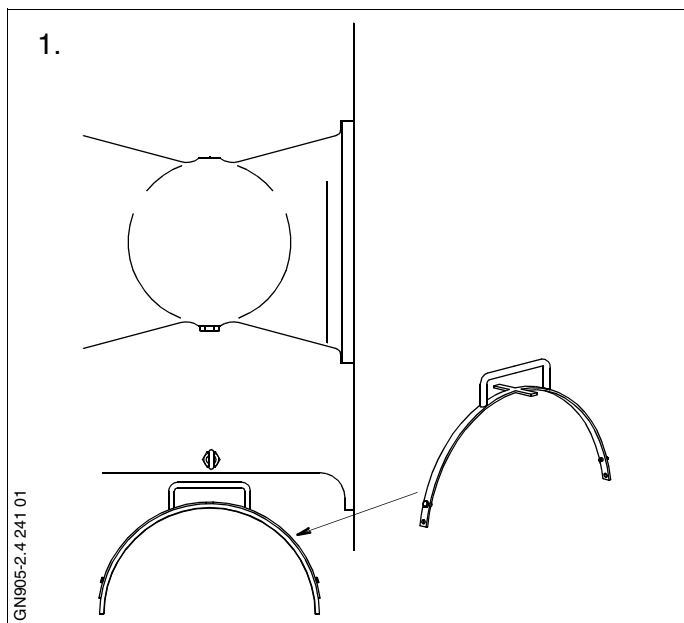
2. Before remounting:

- Check the bearing support for damage and burrs. If damage is found, contact MAN B&W Diesel for advice.
- Make sure that all parts are clean, use non-fluffy cloth to clean between journal and main bearing support.
- Lubricate the bearing journal, the main bearing support and the back of the shell, with main engine lubricating oil.

1 + 2.



KN905-2.2 238 25



Mounting of bearing shells

1. Lubricate the journal and the bearing support with clean oil

Using the bearing shell lifting tool, lift the lower bearing shell into the crankcase and land it on the crankshaft journal.

2. Position the guide tool for the lower bearing shell on the main bearing support. Remove the lifting tool from the bearing shell.

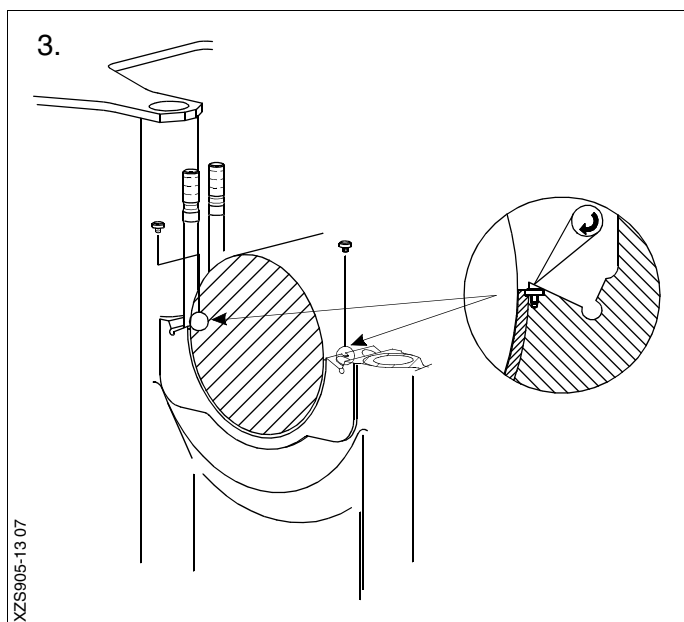
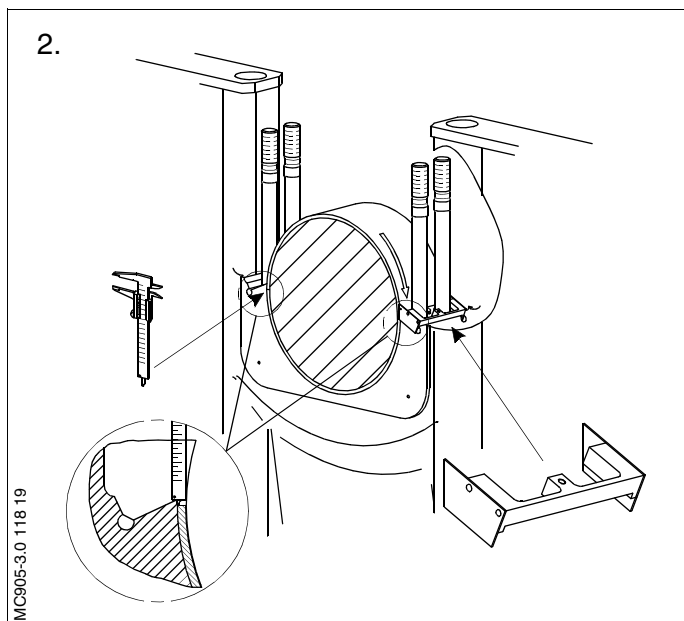
Place the lower bearing shell on the journal and push it down by hand.

To ensure that the shell is correctly mounted, check that the distance from the main bearing support to the shell is equal in both sides.

3. Mount the locking screws in the bearing support.

Lower the crankshaft and remove the hydraulic jacks and the crosspiece from below the crankshaft.

Wipe the contact surfaces between the upper and the lower bearing shells clean with a non-fluffy piece of cloth.



4. Land the upper bearing shell on the main journal in the same way as the lower bearing shell.

Make sure that none of the edges of the upper bearing shell are resting on the edge of the bearing support.

If necessary, place the special tool between the studs and press the shell into place.

Clean the contact surfaces between the bearing support and the bearing cap. Make sure that no oil is left on the contact surfaces.

5. Mount the lifting tool on the bearing cap.

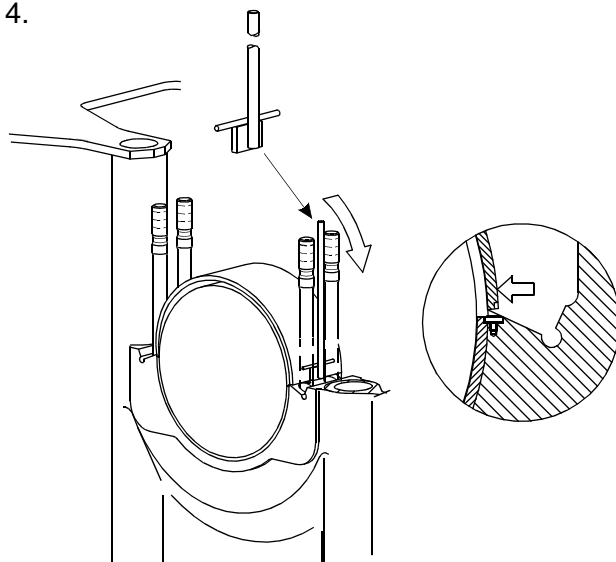
Using the tackles and pulleys, lift the cap into the crankcase in reverse order to dismantling.

See Procedure 905-2.2, steps 12-9.

Note!

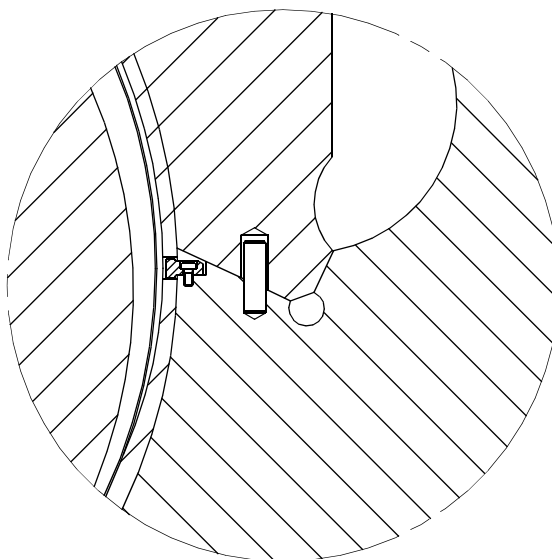
Be sure that the cap lands correctly and that the spring pin in the assembly surface enters the hole in the bearing cap.

4.



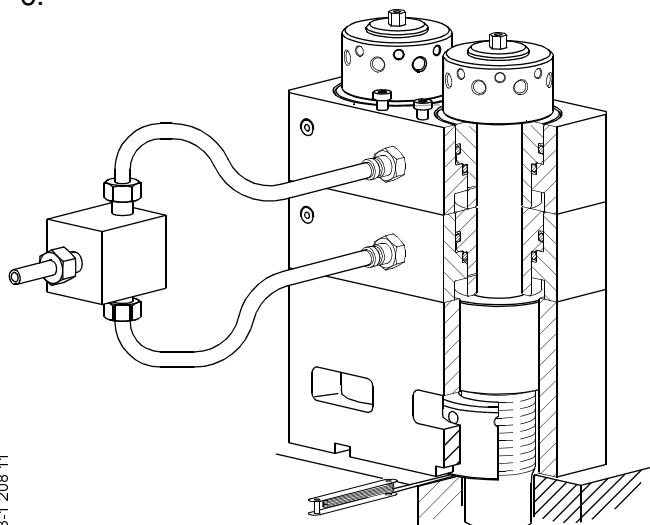
XZS905-13 09

5.



KN905-2.4 238 09

6.



6. Mount the rectangular nut washers and screw the round nuts on to the main bearing studs.

Using the hydraulic jacks, fully tighten the nuts on both sides of the bearing cap.

For tightening pressure, see *Data*.

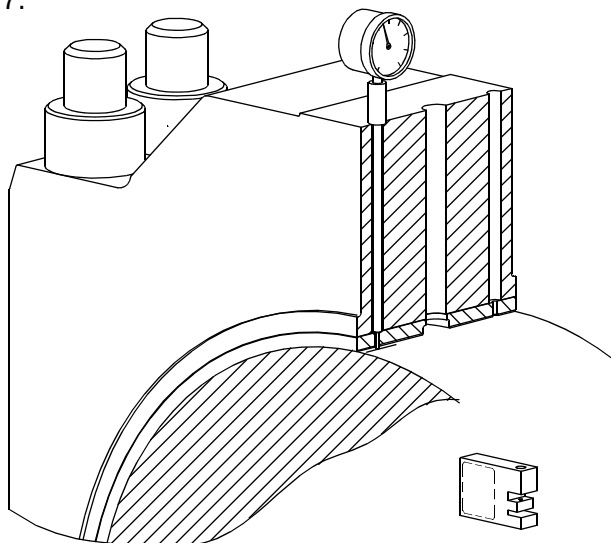
For operation of the hydraulic tools, see *Procedure 913-1*.

7.

Note!

Before removing the hydraulic jacks, check the top clearance between the upper bearing shell and the journal.

7.

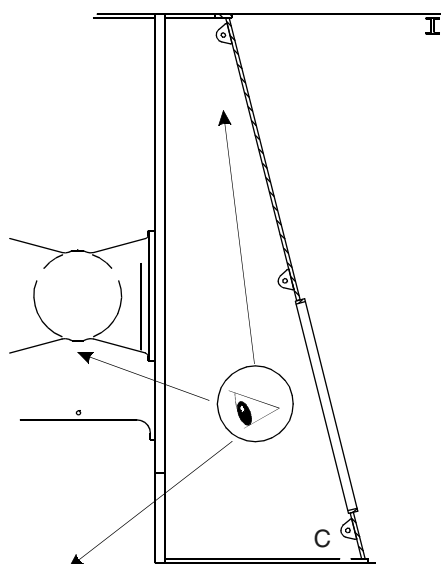


8. Remove all tools from the engine.

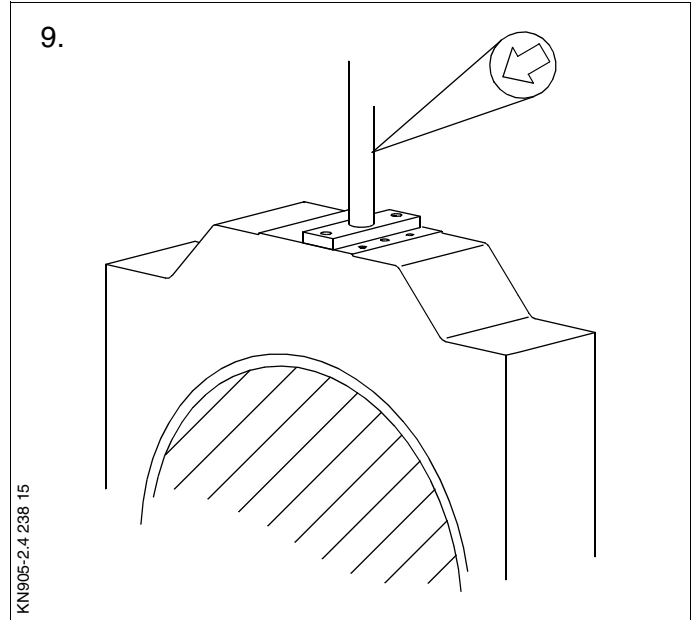
Note!

Search the crankcase to ensure that there are no tools, shackles or rags left behind.

8.



9. Mount the lubricating oil pipe on the main bearing cap.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☐	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D05-06	Thrust bearing segment	27	kg
D05-07	Segment stopper	25	kg
D05-09	Segment stopper, fore	25	kg
D05-10	Segment stopper, aft	18	kg
D05-32	Segment stopper stud, screwing in torque	450	Nm
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400-1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90551	100	Tool for turning out segments
P90662		Camshaft - Hydraulic Tools
P91351	010	Hydraulic pump, pneumatically operated
P91351	046	Hose with unions (1500 mm), complete
P91351	058	Hose with unions (3000 mm), complete
P91351	105	3-way distributor block, complete
P91356		Lifting Tools, Etc.
P91366	058	Feeler gauge

The clearance in the thrust bearing was measured during testbed trials of the engine and noted down in the 'Adjustment Sheet' which is inserted at the front of Volume I, OPERATION.

For a new engine the clearance is 0.5-1.0 mm, and for an engine in service it must not exceed 2.0 mm.

In service it is only necessary to measure the wear of the thrust bearing pads, and to inspect for white metal below the thrust bearing.

1. To measure the wear in the thrust bearing, it is necessary to dismount the foremost segment stopper.

For dismantling the segment stopper, see Procedure 905-3.2.

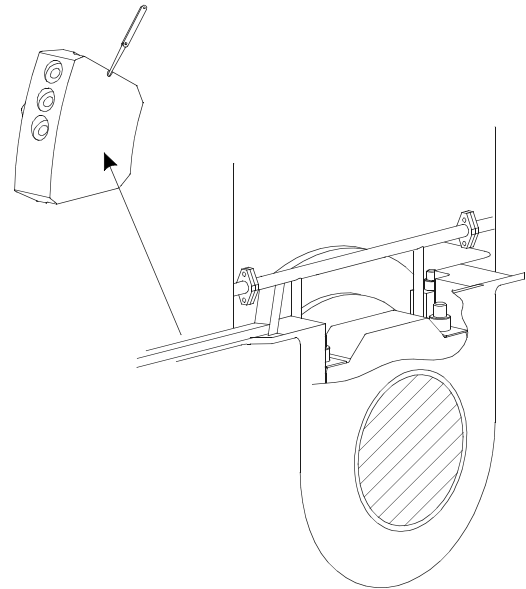
A wear groove of 1 mm is positioned in the uppermost thrust segment. (The segment with thermometers).

2. To measure the wear, the thrust segment has to be pressed against the thrust cam, to eliminate any gap, i.e. by using a suitable crowbar on the back of the segment.

If a feeler gauge of 0.1 mm is **not** able to enter the groove (the wear is more than 0.9 mm), the thrust bearing must be overhauled.

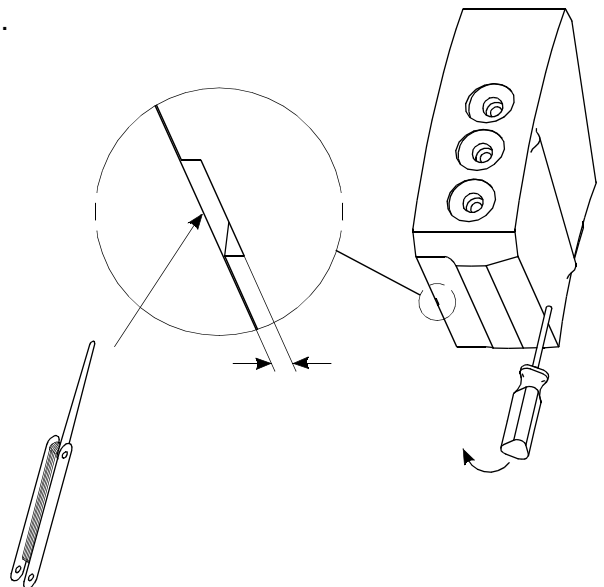
3. Note down the wear for later reference.

1.



GN905-3.1 233 01

2.

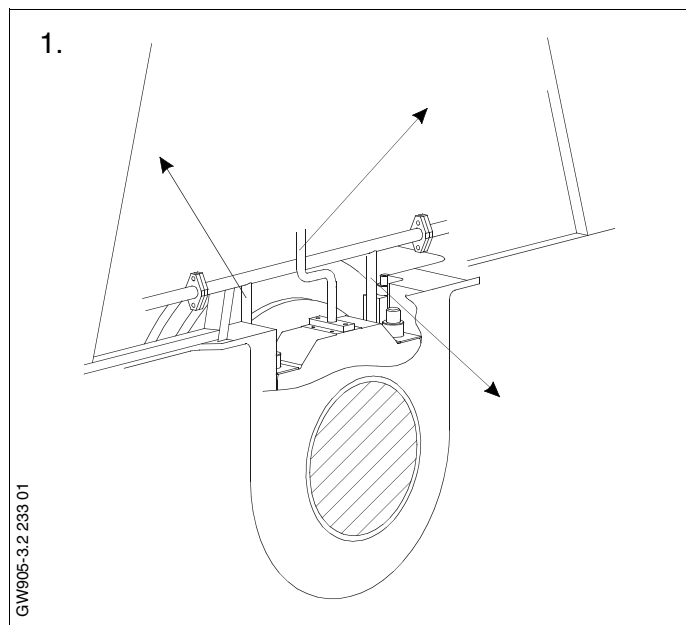


GN905-3.1 224 02

3.

AUTOLOG + CLEARANCES									
CYL 1									
CYL 2									
CYL 3	0,0	0,0	0,0	0,0	0,0				
CYL 4									
CYL 5									
CYL 6									
CYL 7									
CYL 8									

KN905-2.2 238 02



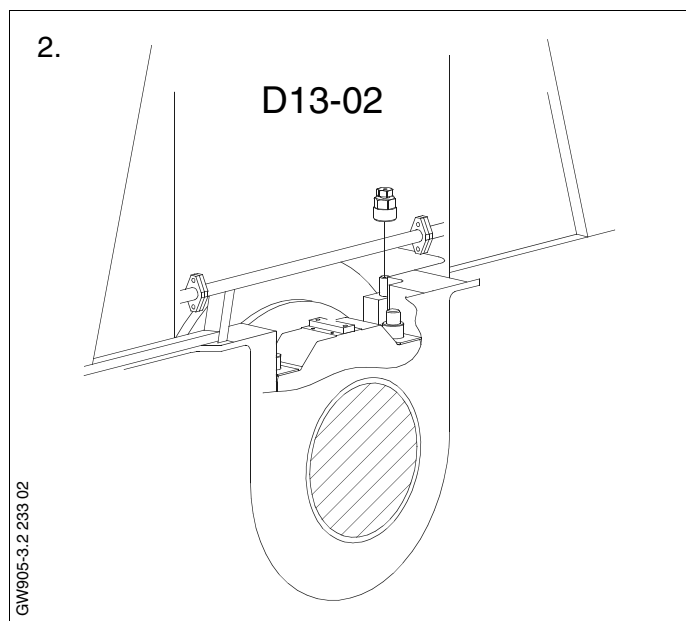
1. Remove the lub. oil pipe from the bearing, the protective shield, and the lub. oil pipe for the thrust segments.

2. Loosen and remove the nuts from the segment stoppers.

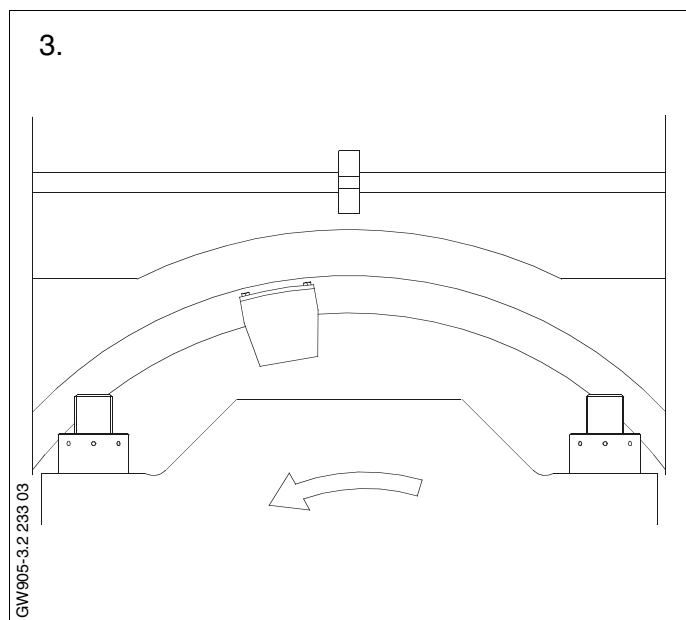
For operation of the hydraulic jacks, see Procedure 913-1.

Use a stud setter to loosen the segment stopper studs. Remove the segment stopper studs.

Mount an eye bolt in the segment stopper and use a rope around the horizontal lub. oil pipe to remove the segment stoppers. Lift the stoppers out of the chain casing.



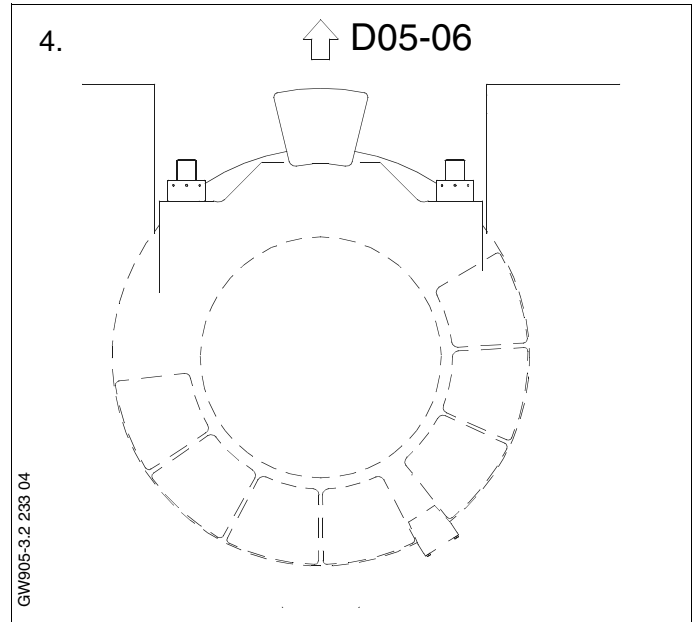
3. Mount the tool for turning out the segments on the edge on the thrust bearing cam.

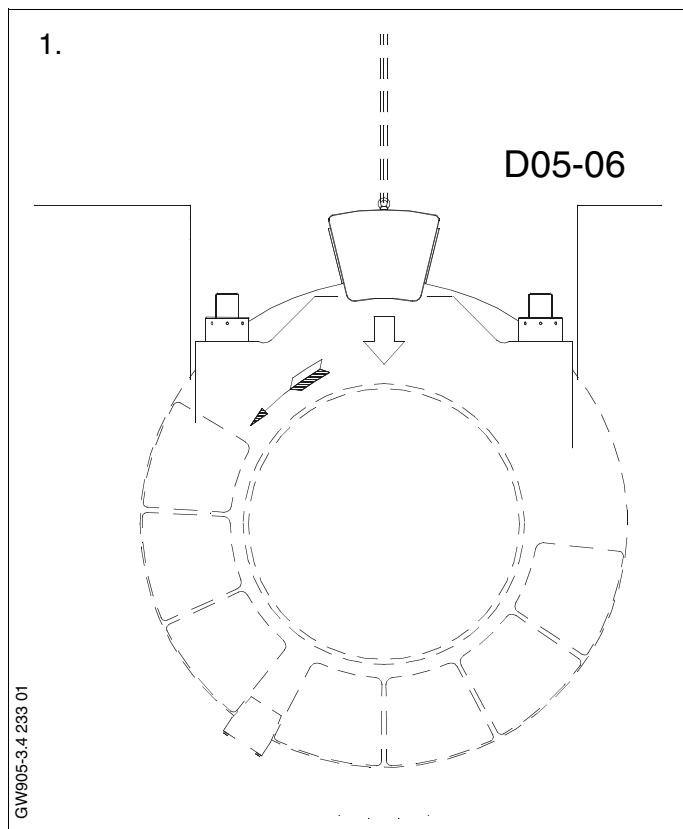


4. Turn up the segments, one at a time, by turning the engine, and remove the segment from the thrust bearing casing. Inspect the segment and remount it before turning up the next segment.
See Procedure 905-3.4.

Note!

Never remove more than one segment at a time.





1. Suspend a tackle above the segments which are to be mounted. Still, it is possible to lift the segments by hand.

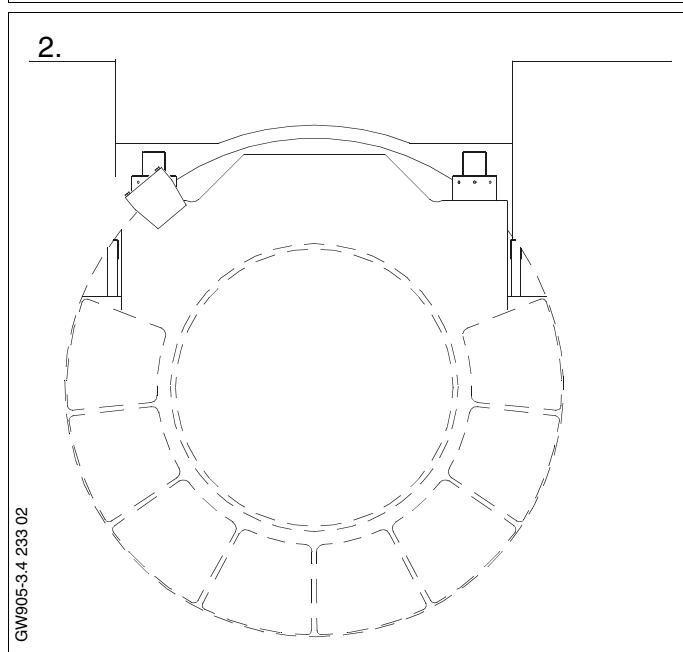
Mount the tool for turning out segments on the thrust bearing cam.

Land the segment on the journal and remove the eye bolt from the segment.

Carefully slide the segment on to the segment tool. Turn the engine to dismantle the next segment.

See Procedure 905-3.2.

2. After the last segment has been mounted, turn the segment tool up and dismount it.



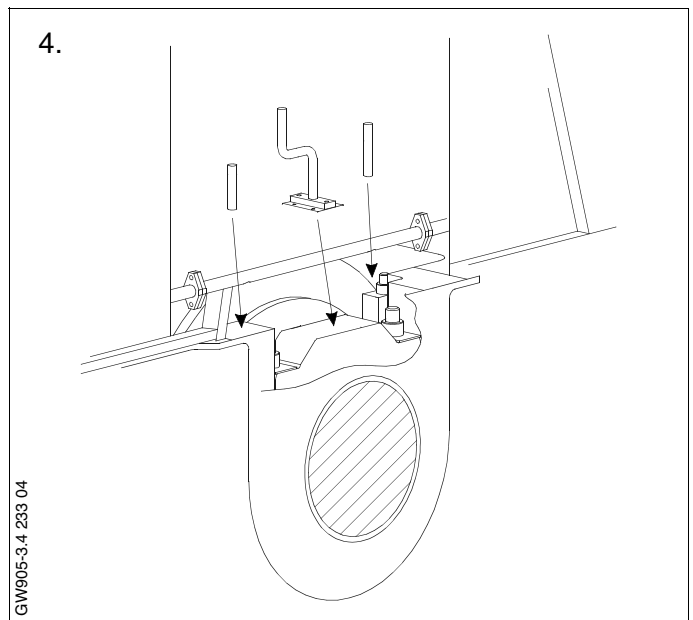
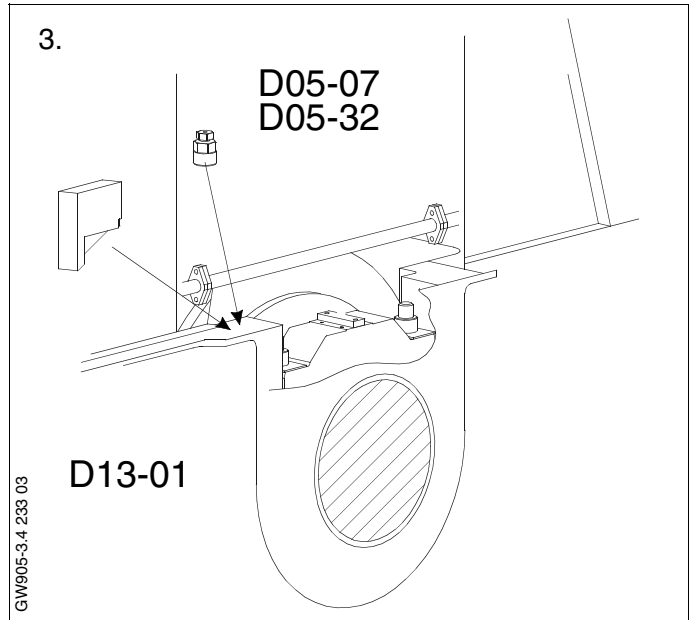
3. Mount the segment stopper.

Mount the segment stopper studs.

Mount the hydraulic jacks and tighten the nuts on the stopper.

For operating hydraulic jacks, see *Procedure 913-1*.

4. Mount the lub. oil pipe for the thrust segments and the lub. oil pipe on the bearing.



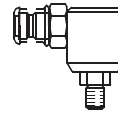
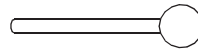
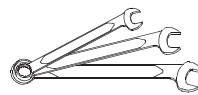
SAFETY PRECAUTIONS

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
X	Engage turning gear
	Shut off cooling water
	Shut off fuel oil
X	Shut off lubricating oil
	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D-1	Min. bearing clearance	0.30	mm
D-2	Max. bearing clearance	0.65	mm
D-3	Main bearing cap	290	kg
D-4	Upper bearing shell	20	kg
D-5	Lower bearing shell	20	kg
D-6	Hydraulic pressure – dismantling	1500-1650	bar
D-7	Hydraulic pressure – tightening	1500	bar

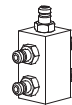
Standard Tools: See Section 913



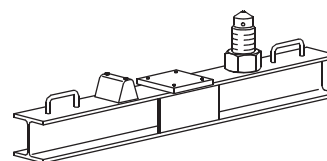
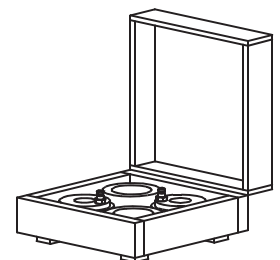
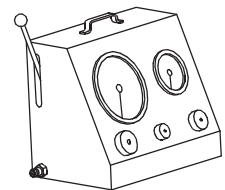
x 2



2 x 3 m



2 x 500 kg



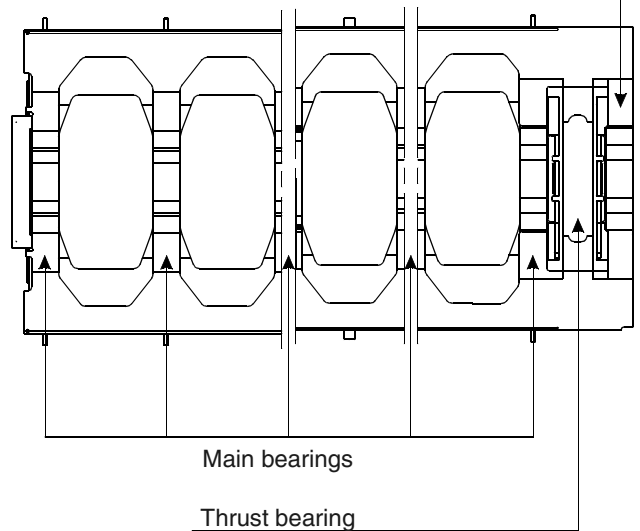


Spare Parts

Plate – Item No.	Description	Qty
91210-xxx 108	Screw	2
91210-xxx 121	Main bearing	1

1. Before any dismantling of the journal bearing, check the clearances as described in *Procedure 905-1.1*.
2. Write down the journal bearing clearances.
3. Also check the journal bearing by visually searching the area underneath the bearing support.

1. Aftmost main bearing (= Journal bearing)



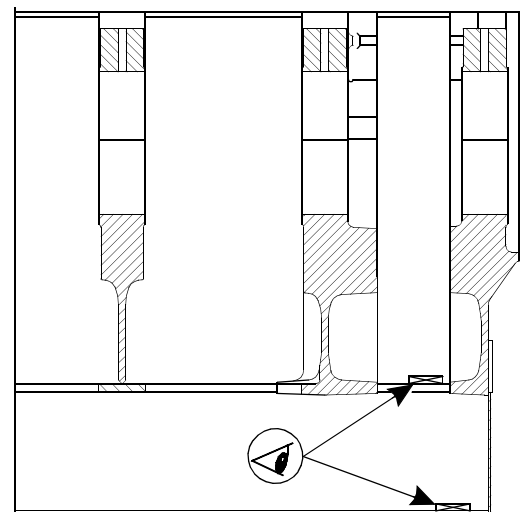
KN905-2.2 238 01

2.

AUTOLOG + CLEARANCES									
CYL 1									
CYL 2									
CYL 3	0,0	0,0	0,0	0,0	0,0				
CYL 4									
CYL 5									
CYL 6									
CYL 7									
CYL 8									

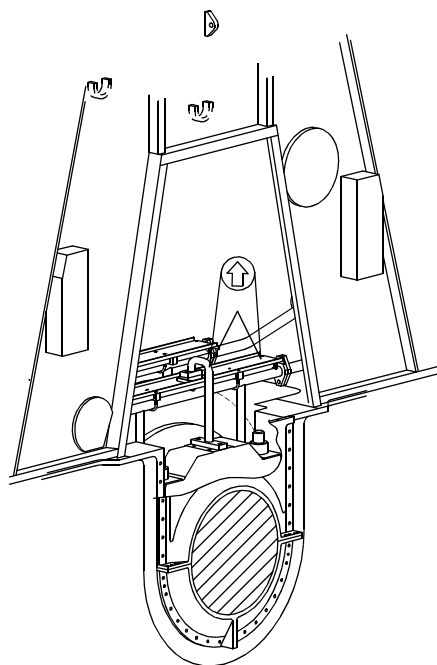
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3.



KN905-4.1 202 03

2.



1. Dismount the screws for the end cover.

Mount a ½-ton tackle above the end cover.

Remove the end cover.

2. Disconnect the lubricating oil pipe from the main pipe.

Remove the protective shield for the thrust bearing.

Turn the crank throw for the aftmost cylinder to TDC.

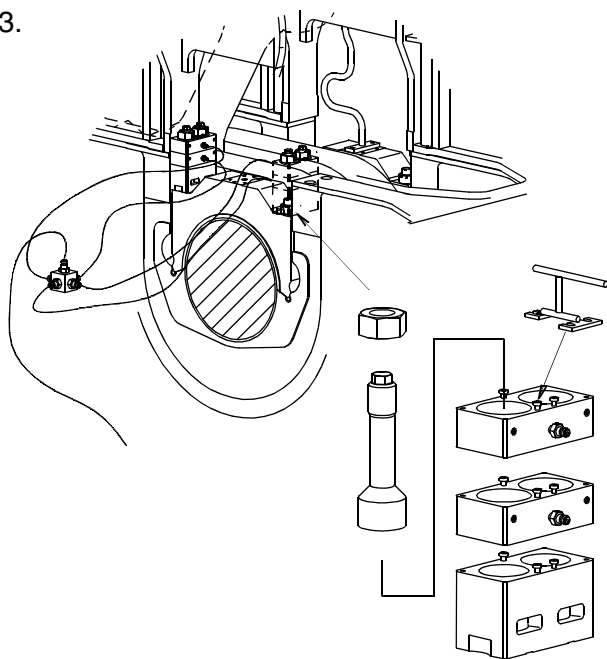
3. Mount the extension studs, the spacers, the hydraulic jacks and the nuts on the extension studs. Loosen the bearing stud nuts, using the hydraulic jacks

For operation of the hydraulic tools, see Procedure 913-1.

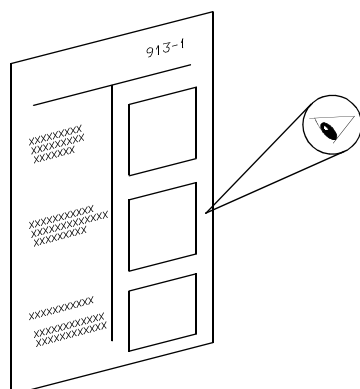
4. Remove the hydraulic main bearing tools from the engine

Unscrew and remove the nuts from the bearing studs.

3.



4.



5. Remove the rectangular nut washers from the bearing cap.

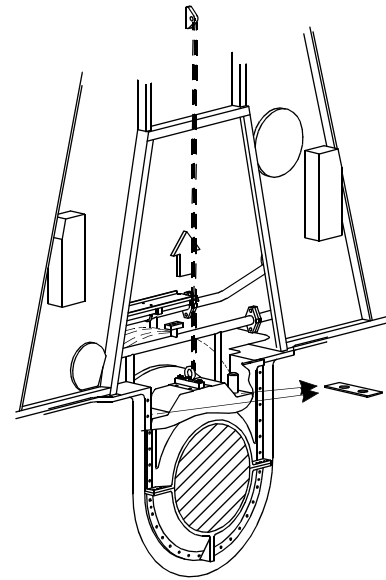
Mount the lifting attachment at the journal bearing cap, and lift the cap out of the engine by using the ½-ton tackle above the bearing.

6. Mount the lifting attachment on the upper shell, and lift the shell out of the engine.
7. Place the crosspiece under the turning wheel as shown. Mount a dial gauge to measure the lift of the crankshaft.

Tighten the screw against the turning wheel and compress the jacks.

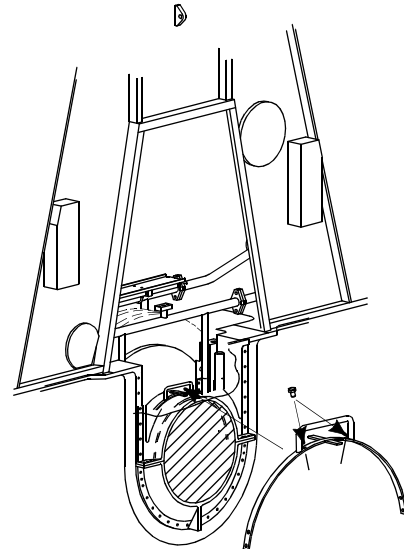
Connect the hydraulic jacks to the high-pressure pump, and raise the pressure until the crankshaft has been lifted 0.2-0.5 mm, but max. the clearance in the adjacent bearing.

5.



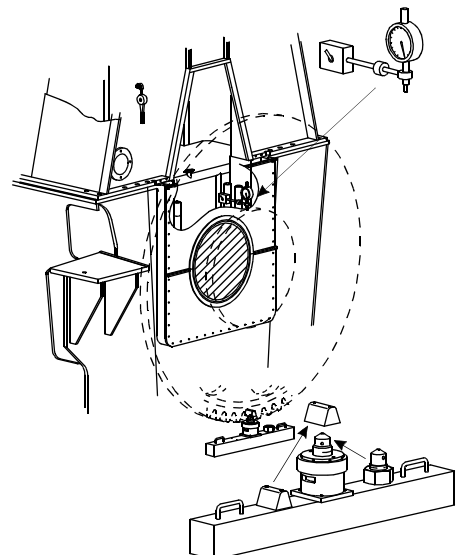
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6.



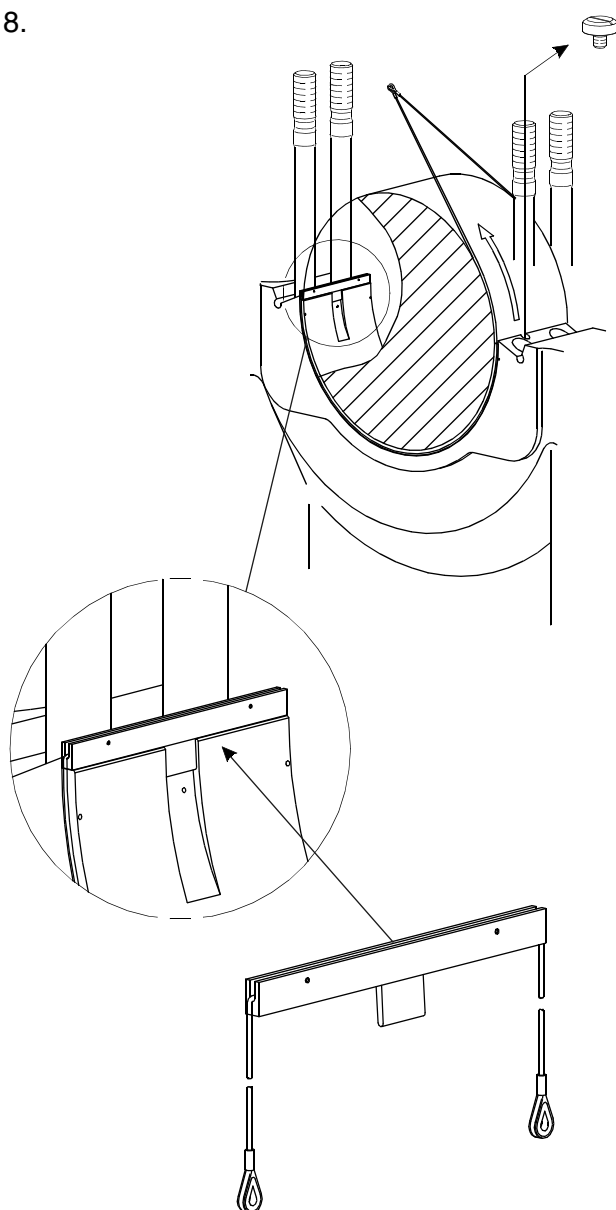
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7.



GN905-2.2 224 19

8.



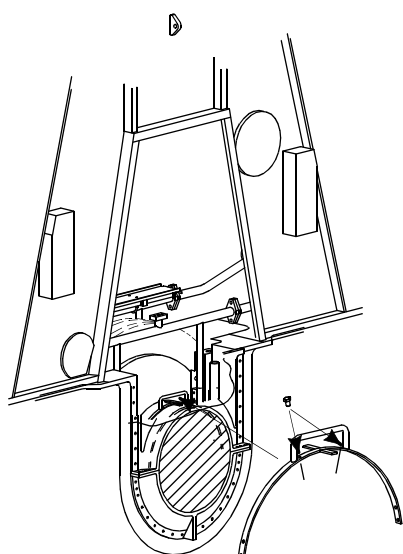
8. Dismount the locking screws.

Place the dismantling tool on top of the lower bearing shell. Make sure that the flap on the dismantling tool enters the oil groove in the bearing shell.

Pull the lower bearing shell round and upwards until it lies over the bearing journal. Never use a tackle larger than 0.5 ton.

9. Take the lower shell out of the engine in the same way as the upper shell, i.e. using the lifting tool.

9.



KN905-2.2 238 26

GN905-2.2 224 17

1. Clean and inspect the bearing shells. It is recommended that the main bearing shells be replaced in pairs.

If it is nevertheless desired to replace only one shell, this requires a careful evaluation of the condition of the shell that is to be re-installed.

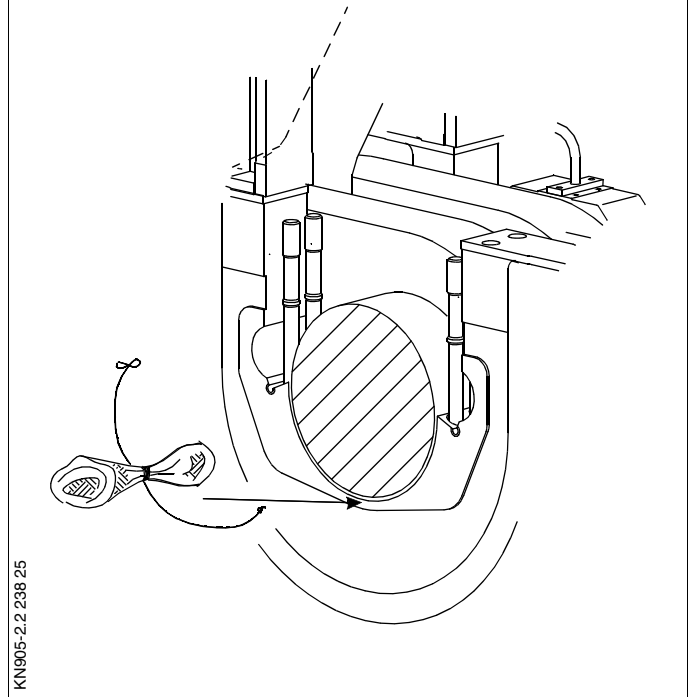
See Instruction book, Volume I, Chapter 708.

For advice on replacing individual bearing shells, it is recommended to contact MAN B&W Diesel A/S or the engine builder.

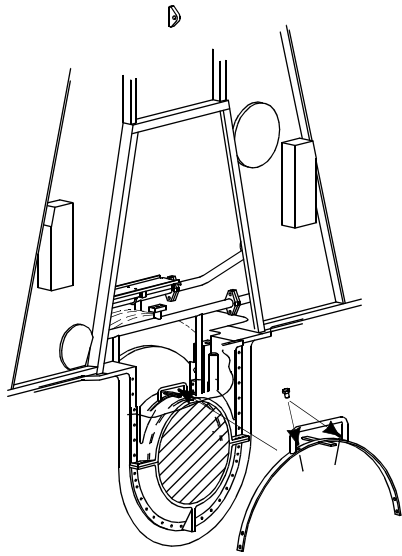
2. Before remounting:

- Check the bearing support for damage and burrs. If damage is found, contact MAN B&W Diesel for advice.
- Make sure that all parts are clean, use non-fluffy cloth to clean between journal and bearing support.
- Lubricate the bearing journal, the bearing support and the back of the shell, with main engine lubricating oil.

1 + 2.



1.



1. Lubricate the journal, the lower bearing shell and the bearing support with clean oil.

Using the bearing shell lifting tool, lift the lower bearing shell into the crankcase and land it on the crankshaft journal.

2. Position the guide tool for the lower bearing shell on the bearing support. Remove the lifting tool from the bearing shell.

Lubricate the journal, the lower bearing shell and the bearing support with clean oil.

Place the lower bearing shell on the journal and push it down by hand.

To ensure that the shell is correctly mounted, check that the distance from the bearing support to the shell is equal in both sides.

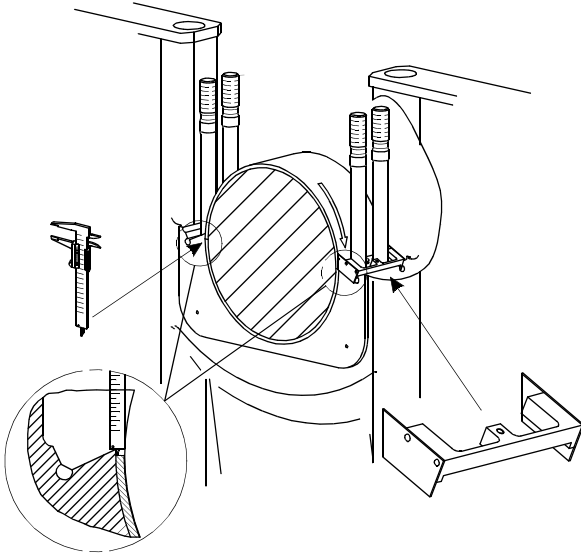
3. Remove the guide tool when the lower bearing shell has been mounted.

Mount the locking screws in the bearing support.

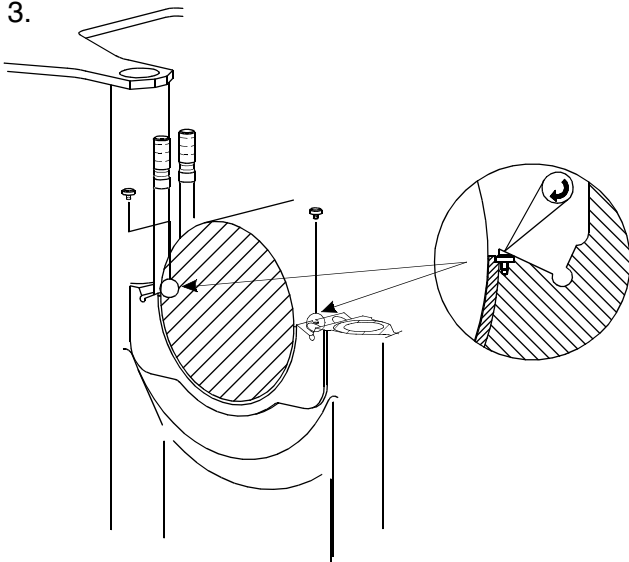
Lower the crankshaft and remove the hydraulic jacks and the crosspiece from under the crankshaft.

Wipe the contact surfaces between the upper and the lower bearing shells clean with a non-fluffy piece of cloth.

2.



3.



4. Land the upper bearing shell on the main journal in the same way as the lower bearing shell.

Make sure that none of the edges of the upper bearing shell are resting on the edge of the bearing support.

If necessary, place the special tool between the studs and press the shell into place.

Clean the contact surfaces between the bearing support and the bearing cap. Make sure that no oil is left on the contact surfaces.

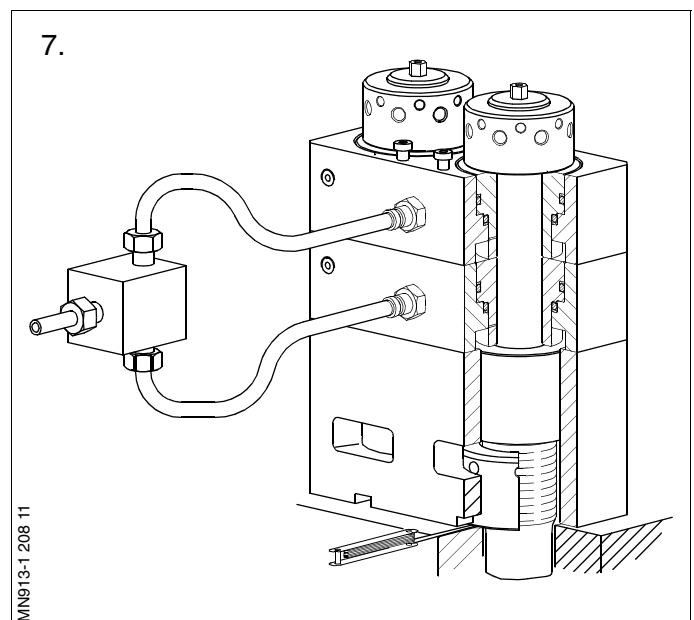
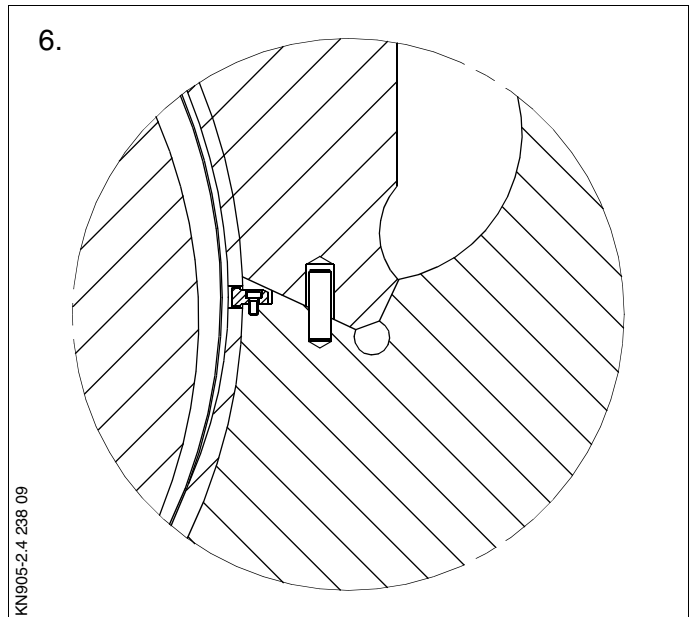
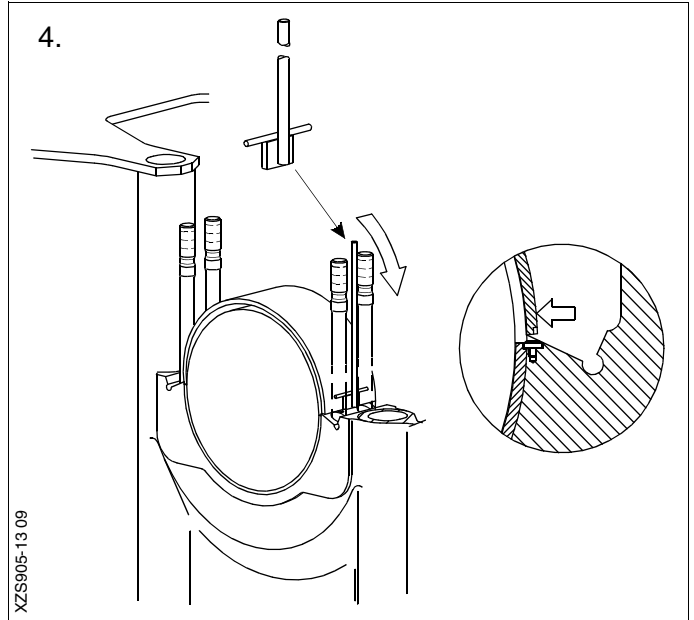
5. Mount the lifting tool on the bearing cap. Using the lifting tool and the tackles, lift the bearing cap into the crank casing in reverse order to dismantling.
See Procedure 905-4, step 5.

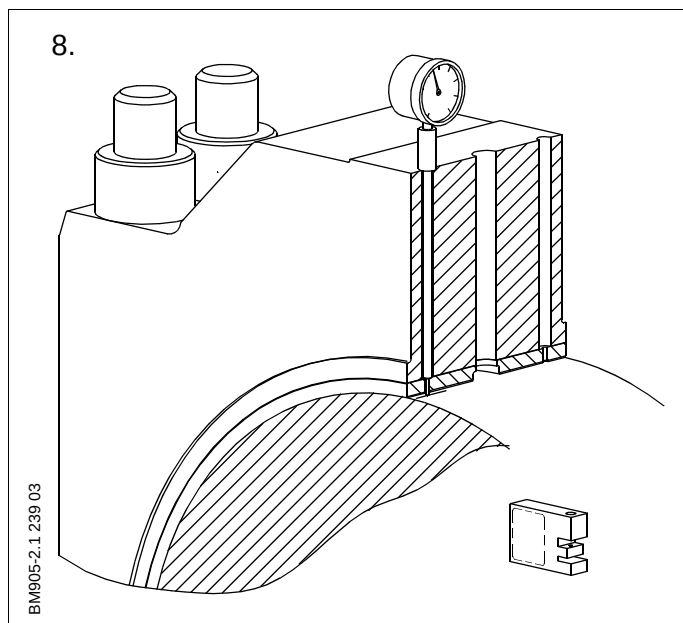
6.

Note!
Be sure that the cap lands correctly and that the guide pin in the assembly surface enters the hole in the bearing cap.

7. Mount the rectangular nut washers, the nuts, the spacer rings, the hydraulic jacks, and tighten the nuts. *See Data.*

For operation of the hydraulic tools, *see Procedure 913-1.*





8.

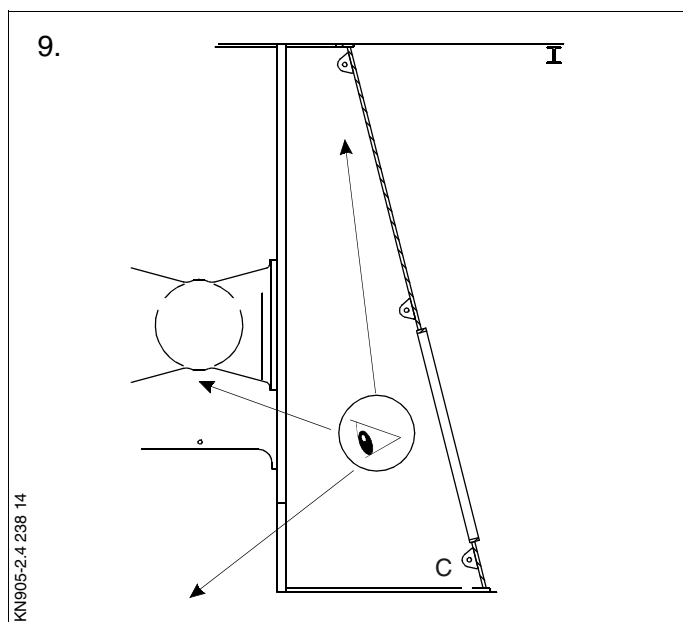
Note!

Before removing the hydraulic jacks, check the top clearance between the upper bearing shell and the journal.

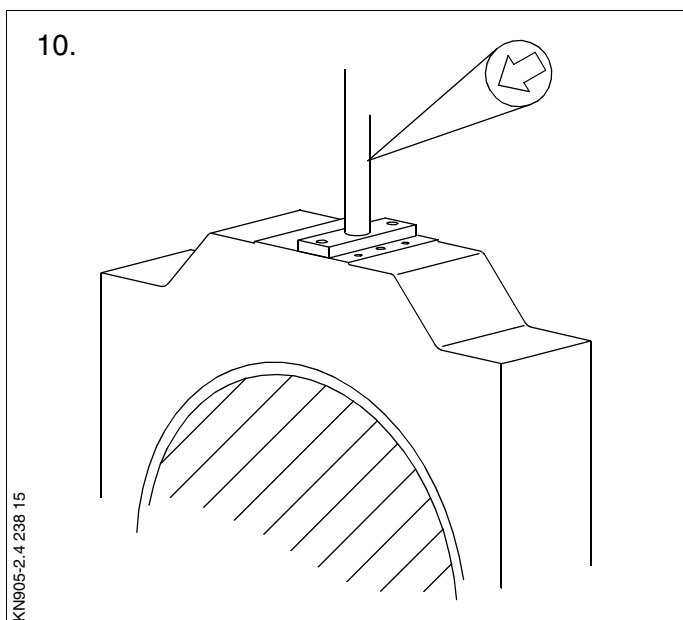
9. Remove all tools from the engine.

Note!

Search the crankcase to ensure that there are no tools, shackles or rags left behind.



10. Mount the lubricating oil pipe on the bearing cap, and mount the end cover.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D05-18	Inner studs, screwing-in torque	150	Nm
D05-19	Nuts on inner studs, tightening torque	500	Nm
D05-22	Outer studs, screwing-in torque	450	Nm
D05-23	Horizontal screws, tightening torque	450	Nm
D05-24	Horizontal screws, tightening torque/angle	100/25	Nm/°
D05-25	Damper housing, upper part	275	kg
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400-1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90662		Camshaft - Hydraulic Tools
P91351	010	Hydraulic pump, pneumatically operated
P91351	058	Hose with unions (3000 mm), complete
P91351	105	3-way distributor block, complete
P91356		Lifting Tools, Etc.
P91359		Torque Spanners

For checking the effectiveness of the axial vibration damper, it is necessary to measure the longitudinal movements of the fore end of the crankshaft during running.

The measurement (or reading) should be taken at the same r/min as during the sea trials. (Preferably 90% and 100% of MCR.)

As different equipment may be mounted on the specific engines, the checking procedure describes three systems:

A: Electronic, with Axial Vibration Monitor

B: Mechanical, (without angle encoder fore)

C: Mechanical, with angle encoder fore.

A: Electronic, with Axial Vibration Monitor

The probe is connected to a control unit which displays peak-to-peak movements and sends signals to the engine control system.

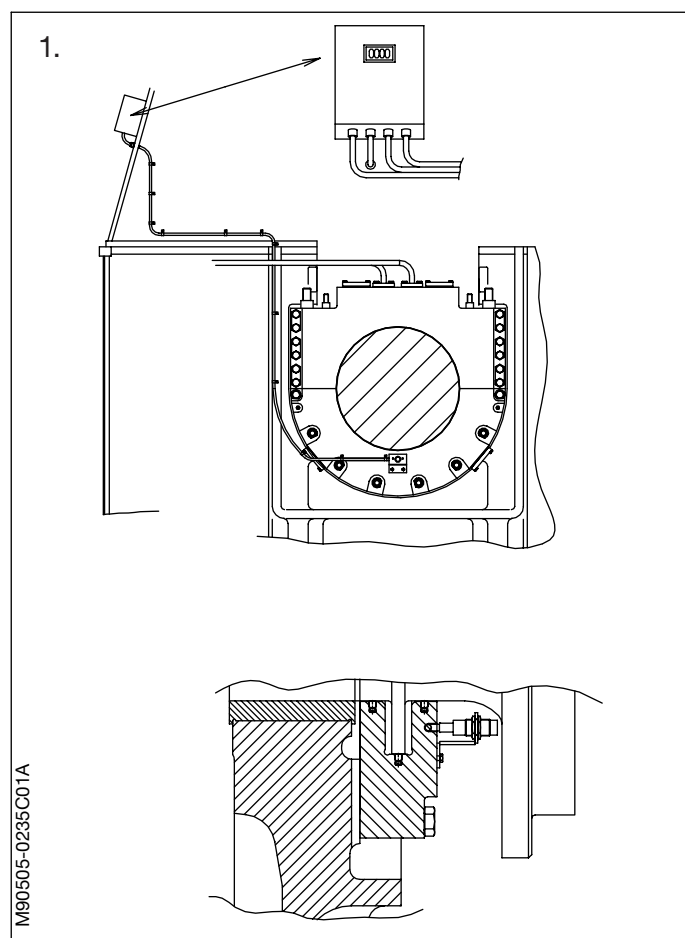
A proximeter probe is built onto the damper housing. The probe can either be fitted on the lower part of the damper housing or on the upper part of the damper housing.

1. In this case, the proximeter probe is built onto the lower part of the damper housing.

Concerning overhaul and setting of the electronic device, see *makers instructions and Volume I, Operation, Chapter 701*.

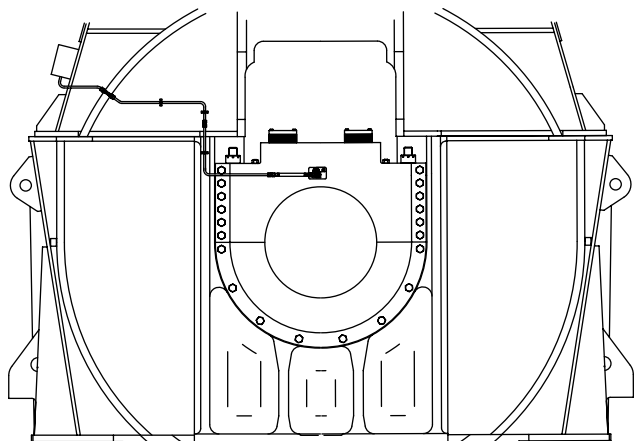
The peak-to-peak values displayed in mm are to be compared with the original values obtained during sea trial and the limits given in *Volume I, Operation, Chapter 701*.

If the peak-to-peak value exceeds the *Normal Service Value*, it is necessary to overhaul the axial vibration damper, see *Procedure 905-5.2*.



2.

M90505-0235C02A



2. In this case, the proximeter probe is mounted on the upper part of the damper housing.

The overhaul and settings of the electronic device are the same as those mentioned under point 1. Interpretation of peak-to-peak displayed values is also the same as mentioned under point 1.

B: Mechanical (without angle encoder fore)

1. Stop the engine.

Unscrew the two plugs from the cover at the front of the engine.

2. Fit the shaft piece to the end of the crankshaft, using the handle.

After tightening the shaft, remove the handle.

Screw the measuring arm into the small threaded hole beside the centre hole, so that the arm is perpendicular to the shaft piece. Tighten the lock nut.

Attach a pencil to the arm.

Note!

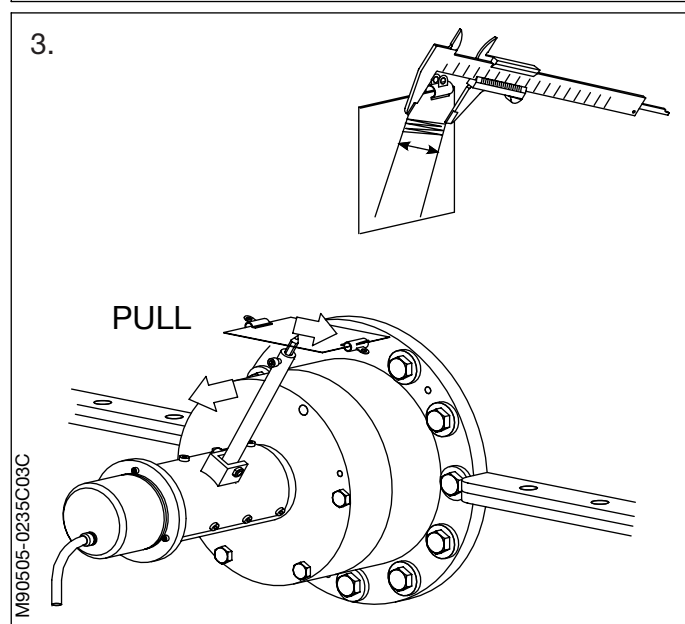
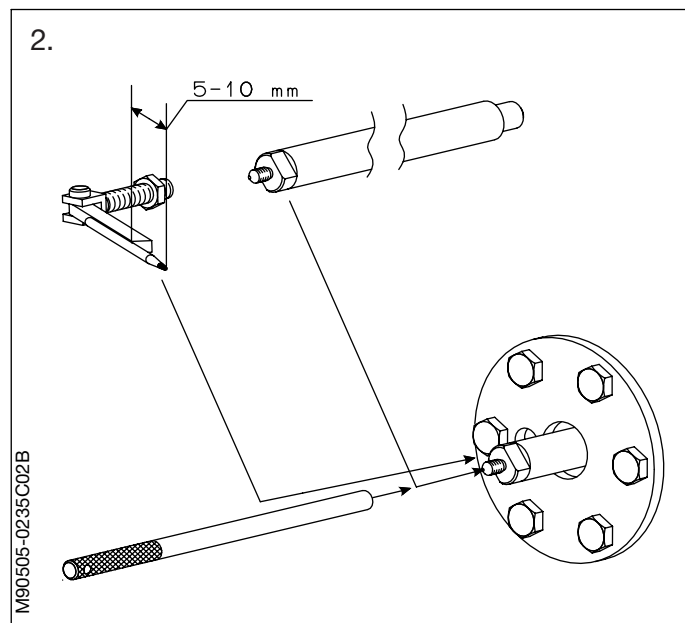
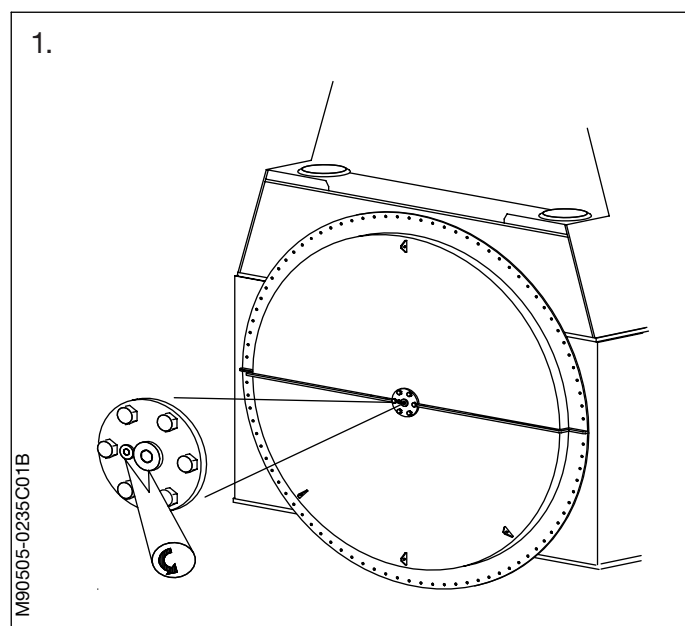
To obtain a correct measurement, the tip of the pencil should protrude 10 mm from the end of the arm.

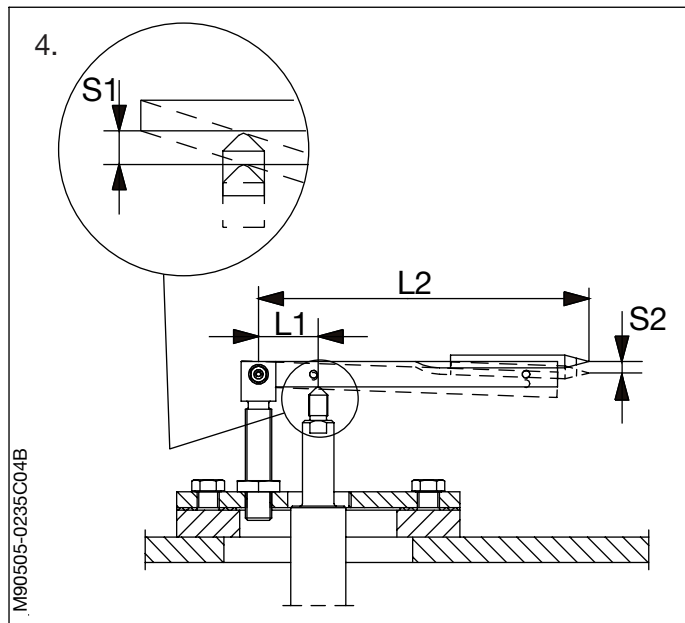
3. Start the engine, and let the speed rise to the number of revolutions at which the measurement is to be taken. (Preferably 90% and 100% of MCR.)

When measuring, press the measuring arm against the shaft piece with the one hand. With the other hand, move a sheet/block of paper, clipped on to a solid backing plate, lightly downwards against the tip of the pencil.

To ensure a certain inertia, the backing plate should have a mass of approx. 1-2 kg.

The axial movements (S2) recorded on the paper must be measured with a slide caliper as shown in the sketch.





4. Before comparing the measured value (S2) with the *Normal Service Value* (S1), stated in *Volume I, Operation, Chapter 701*, it is necessary to compensate for the ratio in the tool.

For a standard measuring tool ($L_1 = 38 \text{ mm}$ and $L_2 = 200 \text{ mm}$), the ratio is 5.3. Therefore, S1 can be calculated as follows:

$$S1 = \frac{S2}{5.3}$$

If a non-standard tool is used, S1 can be calculated as follows:

$$S1 = S2 \times \frac{L_1}{L_2}$$

If the peak-to-peak value exceeds the *Normal Service Value*, it is necessary to overhaul the axial vibration damper, see *Procedure 905-5.2*.

C: Mechanical, with angle encoder fore

For engines with angle encoder fore, a measuring tool is available for mounting on the angle encoder housing.

1. Stop the engine.

Dismantle the protection shield over the angle encoder. (Not always on engine).

Dismantle the cover on the angle encoder housing.

Apply some Molycote™ on the fore side of the shaft cam.

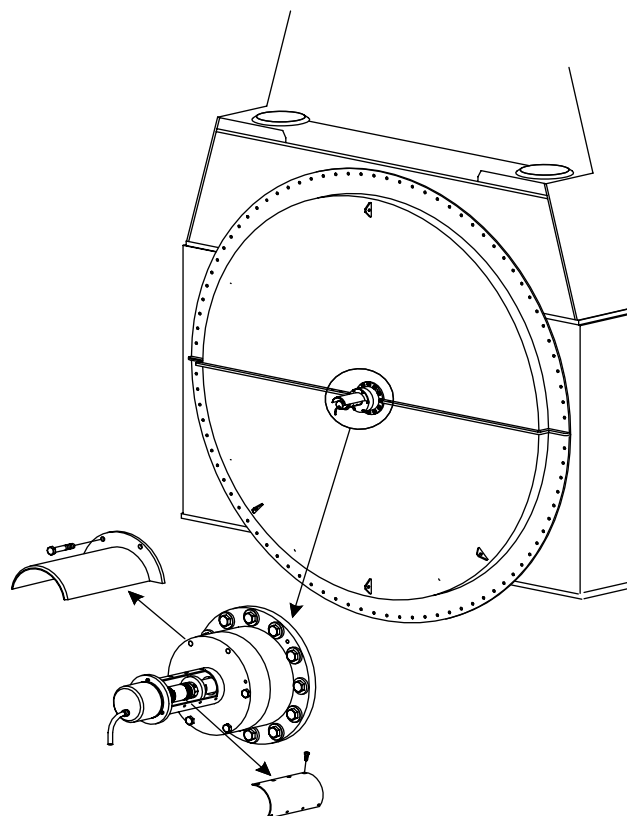
2. Fit the axial vibration measuring tool, using the screws from the cover.

Attach a pencil to the arm.

Note!

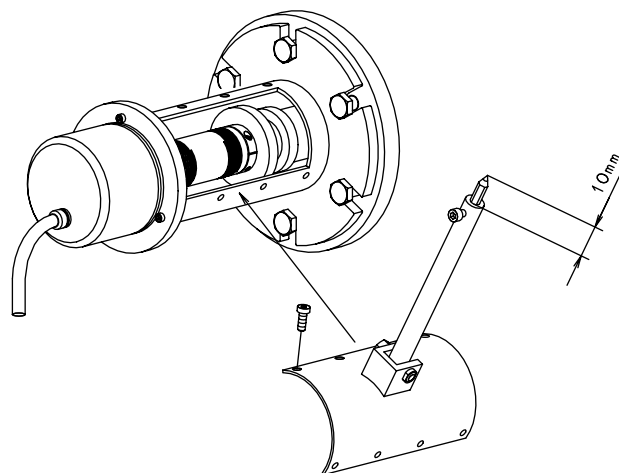
To obtain a correct measurement, the tip of the pencil should protrude 10 mm from the end of the arm.

1.

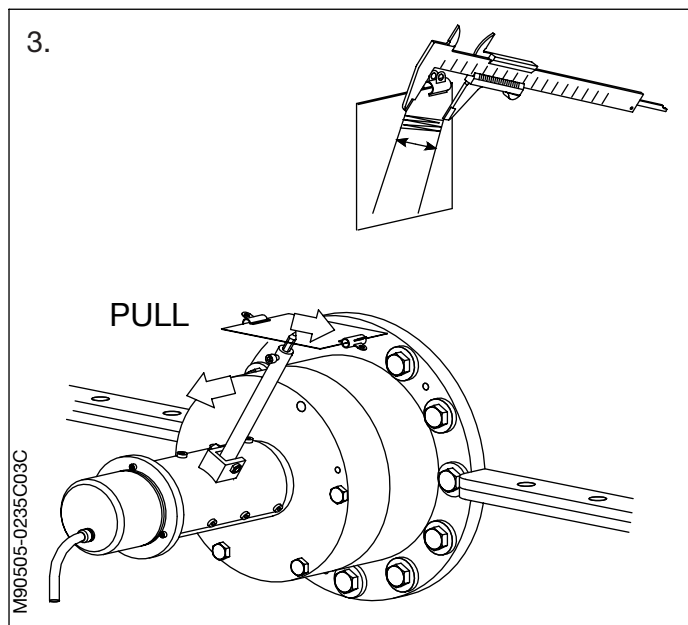


M90505-0235C01C

2.



M90505-0235C02C

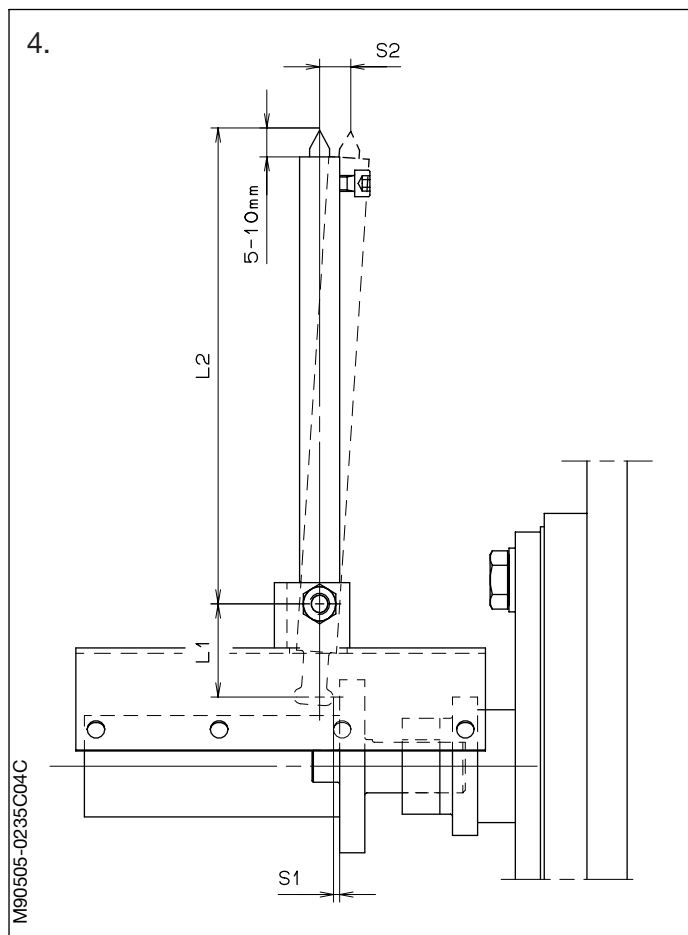


3. Start the engine, and let the speed rise to the number of revolutions at which the measurement is to be taken. (Preferably 90% and 100% of MCR.)

When measuring, pull the measuring arm against the shaft cam with one hand. With the other hand, move a sheet/block of paper, clipped onto a solid backing plate, lightly sideways against the tip of the pencil.

To ensure a certain inertia, the backing plate should have a mass of approx. 1-2 kg.

The axial movements (S2) recorded on the paper must be measured with a slide calliper as shown in the sketch.



4. Before comparing the measured value (S2) with the *Normal Service Value* (S1), stated in *Volume I, Operation, Chapter 701*, it is necessary to compensate for the ratio in the tool.

For a standard measuring tool ($L_1 = 35$ mm and $L_2 = 178$ mm), the ratio is 5.1. Therefore, S1 can be calculated as follows:

$$S1 = \frac{S2}{5.1}$$

If a non-standard tool is used, S1 can be calculated as follows:

$$S1 = S2 \times \frac{L_1}{L_2}$$

If the peak-to-peak value exceeds the *Normal Service Value*, it is necessary to overhaul the axial vibration damper, see *Procedure 905-5.2*.

Dismantling of the axial vibration damper is carried out from the crankcase of cyl. No. 1.

1. Turn the engine to 90° after TDC.

Disconnect the lubricating oil pipes for the main bearing and axial vibration damper.

Loosen and remove the nuts from the inner studs and the hydraulic nuts from the outer studs. *For operation of the hydraulic tools, see Procedure 913-1.*

Loosen and remove the horizontal screws.

Only the upper part of the housing needs to be removed, while the lower half remains mounted on the bedplate.

2. Fit an eyebolt in the centre lifting hole on the upper part.

Engines with chain box fore:

Fit a tackle in the bracket above the axial vibration damper inside the chain box.

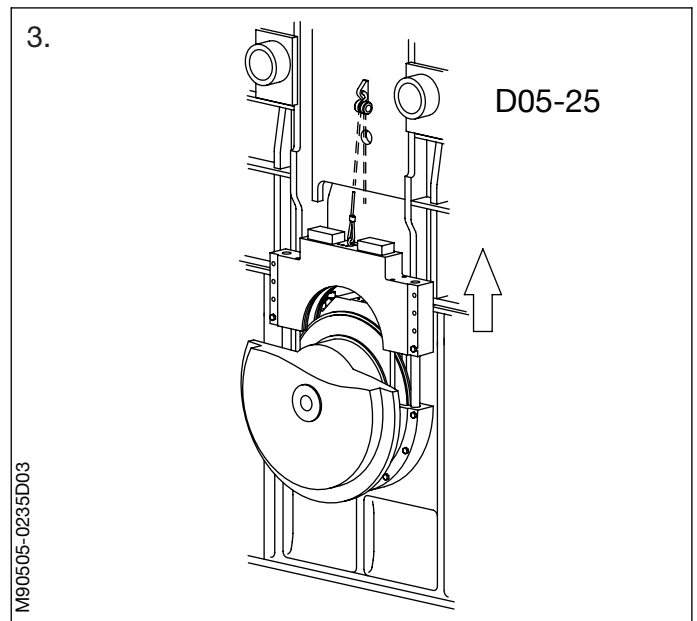
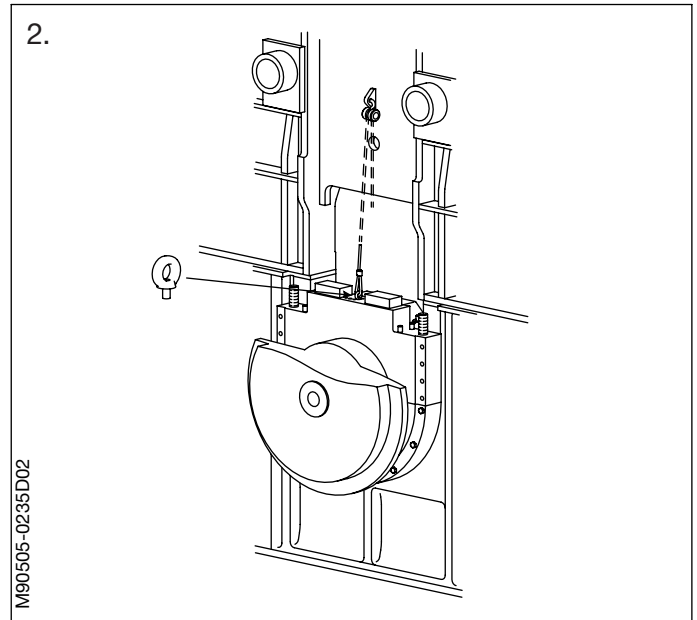
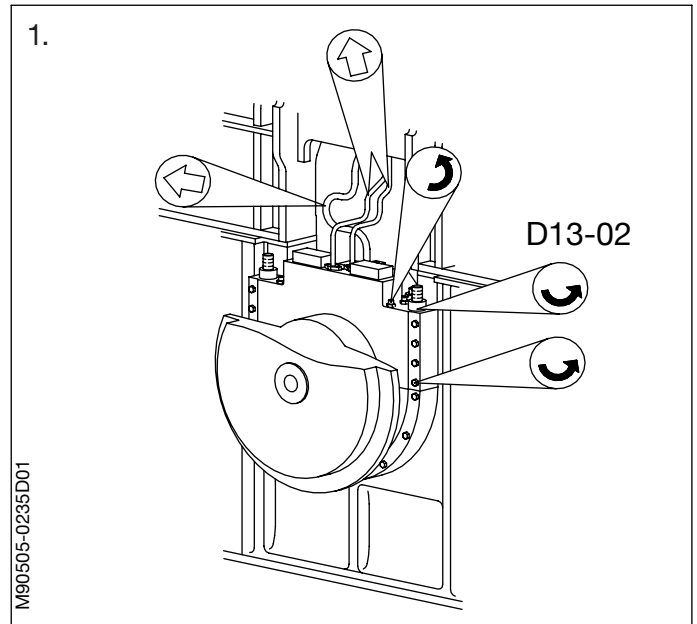
Engines without chainbox fore:

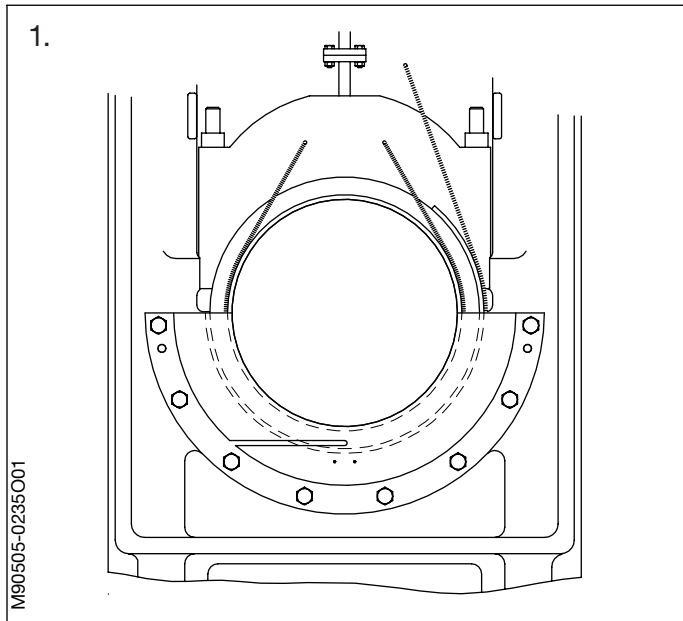
Fit a tackle below the fore end platform, and dismount the cover on the lifting hole above the axial vibration damper.

3. By means of the tackle and wire rope, separate and raise the upper part of the damper housing to a position in which it is possible to change the oil seals and springs.

Note!

Before starting the work under the suspended upper part, secure the upper part with, e.g. a wire rope or another tackle.





1. Dismantle and remove the springs from the oil sealing rings.

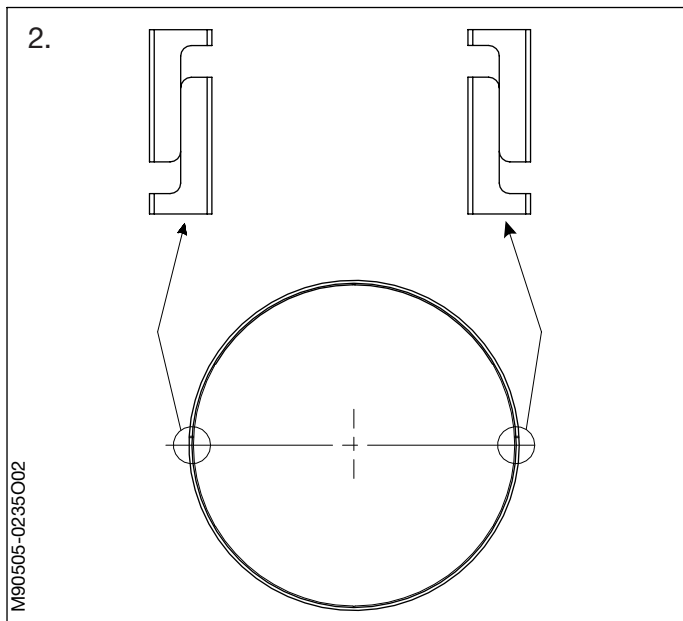
Remove the oil sealing rings.

Note!

It is recommended to discard both the springs and the oil sealing rings.

When installing the new oil sealing rings, first insert the lower half of the rings in the lower housing.

Then press the tension springs into the groove between the oil sealing ring half and the housing.



2. Install all the upper halves of the oil sealing rings in such a way that the clearance at both joints on each oil sealing ring is the same.

Now hook the new tension springs into place, ensuring that they are centralised in the grooves of the oil sealing rings.

Note!

The lower half of the oil sealing rings will protrude above the centreline.

1. Release and carefully lower the upper part of the damper housing.

When reaching the oil sealing rings, take care that the rings enter the sealing ring grooves correctly.

Before landing the upper part on the lower part, be sure that the guide pins have entered the guide pin holes.

2. Install and tighten the vertical nuts. See Data.

Install and tighten the hydraulic nuts. *For operation of the hydraulic tools, see Procedure 913-1.*

Fit and tighten the horizontal screws. See Data.

Note!

Search the crankcase to ensure that there are no tools, shackles or rags left behind.

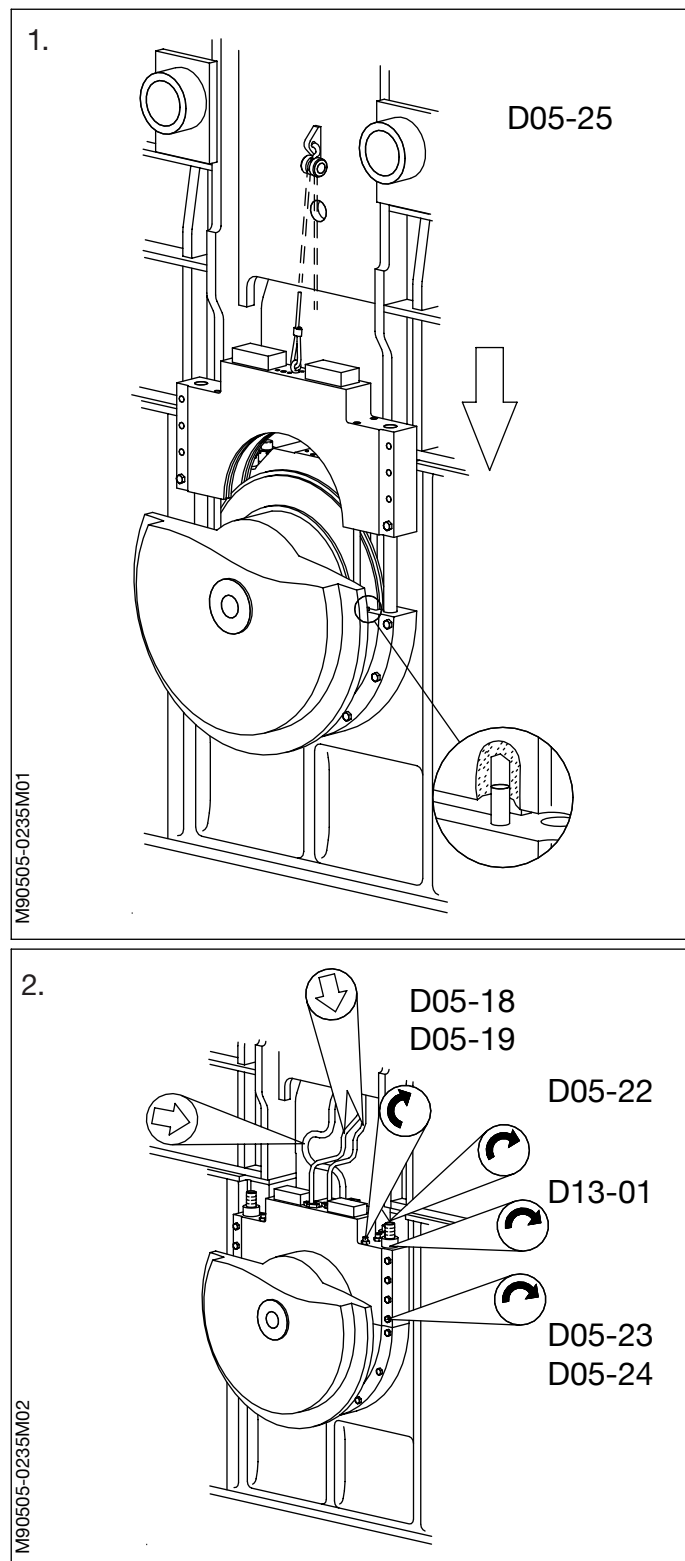
Connect the lubricating oil pipes.

Engines without chainbox fore:

Fit the cover for the lifting hole above the axial vibration damper.

Note!

After overhaul of the axial vibration damper, it is recommended to check the axial vibrations. See *Procedure 905-3.1*.



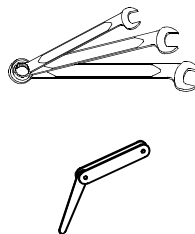
SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D-1	Tacho pick-ups for governor system(Gov. and BMS maker : Kongsberg) – check distance	5	mm

Standard Tools: See Section 913





Spare Parts

Plate – Item No.

Description

Qty

The engine is equipped with below pick-ups:

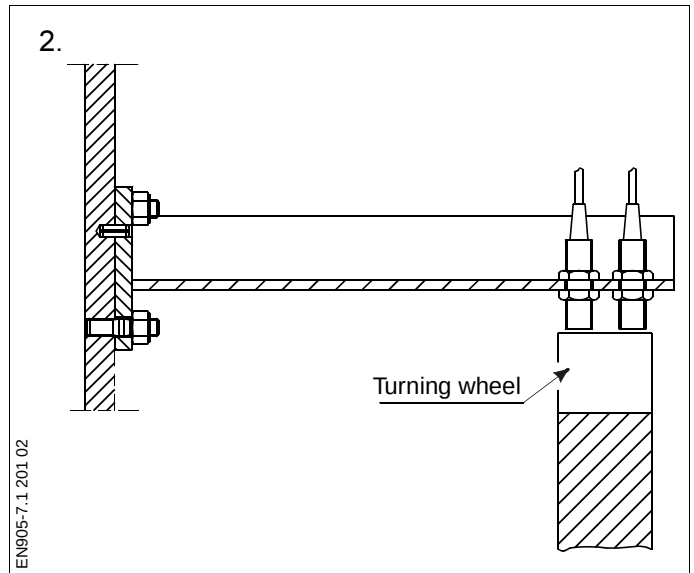
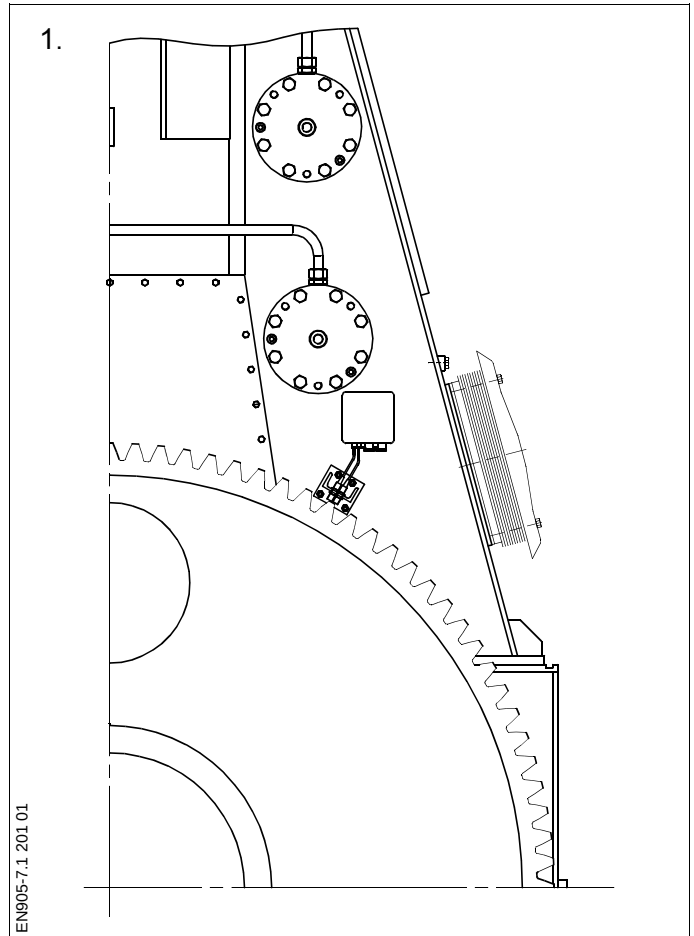
- Governor system pick-ups

Governor system pick-ups:

1. The tacho pick-ups are placed next to the turning wheel.
2. Check the distance between the tacho pick-ups and the turning wheel, see *Data*.

If necessary, adjust the distance.

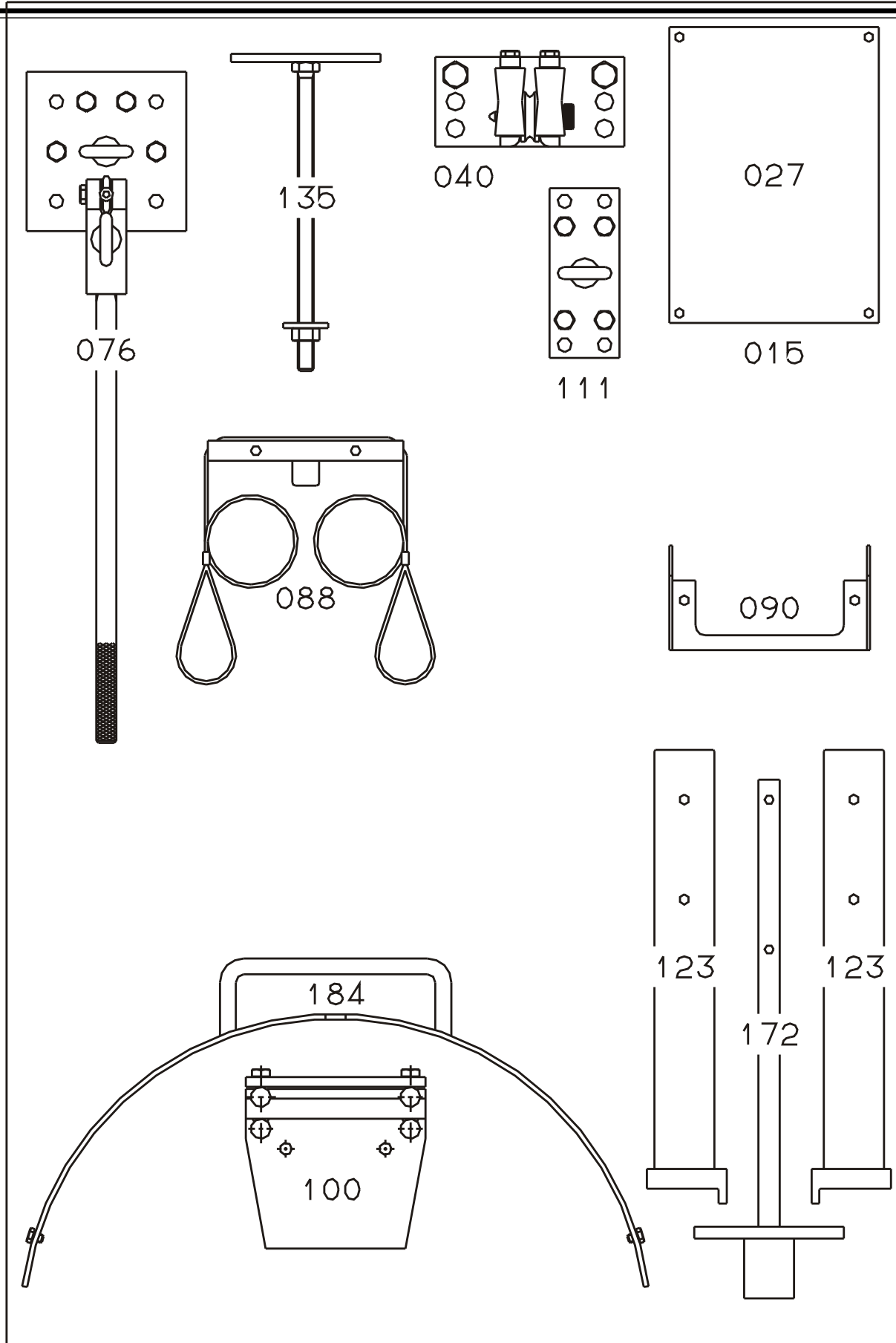
If the tacho pick-ups are renewed, note down the wiring to ensure the correct re-mounting.



Crankshaft, Thrust Shaft, Main Bearing, etc. - Panel

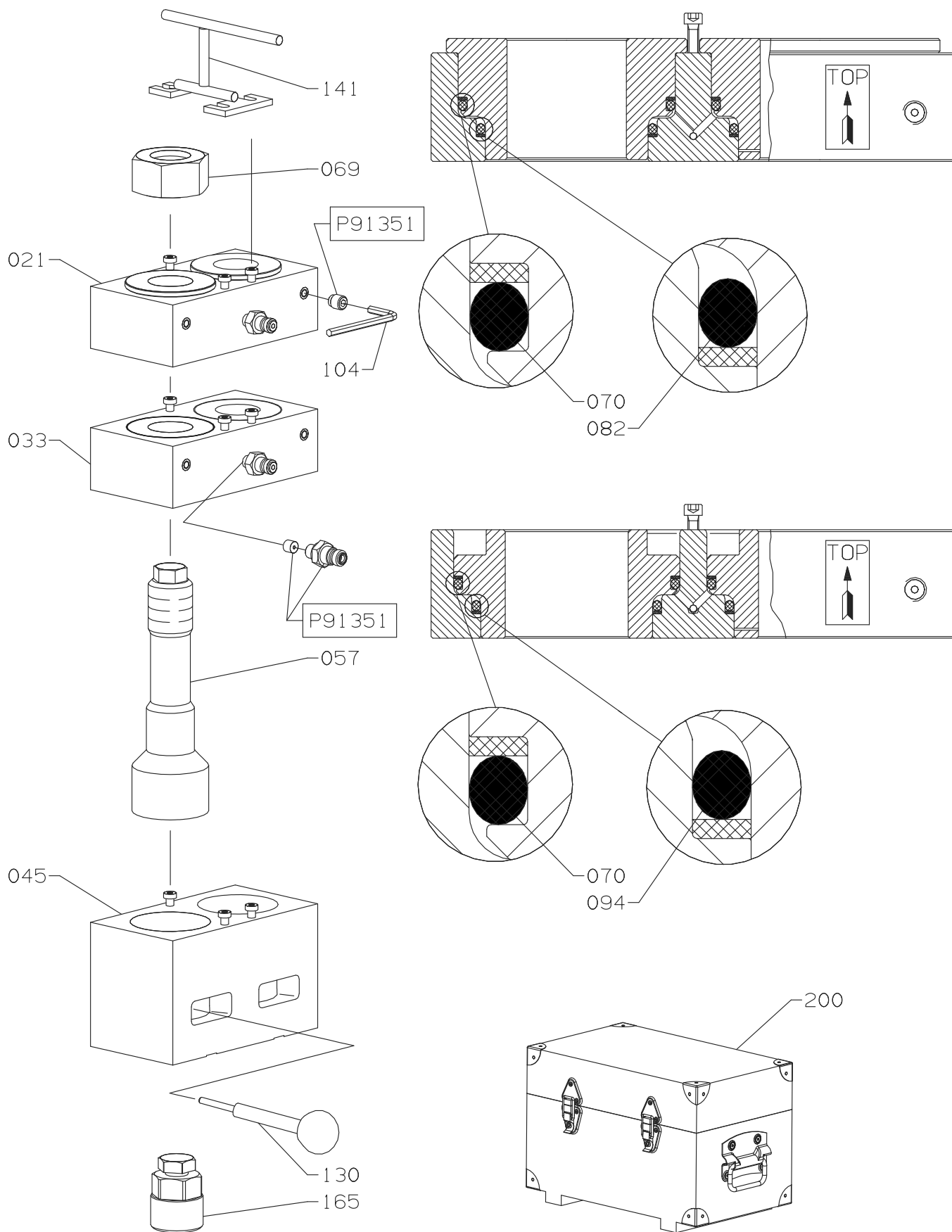
MAN B&W

Plate
P90551-0227

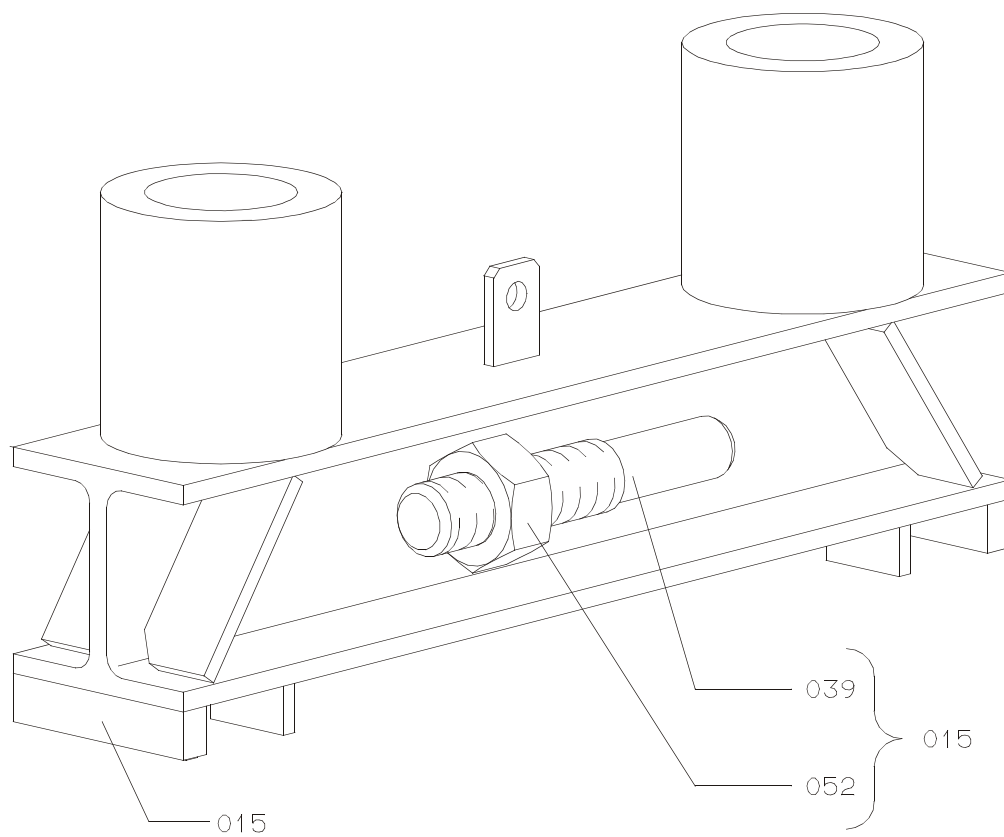


P90551-0227

Item No.	Item Description		
015	Panel for tools		
027	Name plate		
040	Wire pulley, main bearing		
076	Lifting tool, main bearing cap		
088	Dismantling tool, main bearing shell		
090	Mounting tool, thin bearing shell		
100	Tool, turning out segments		
111	Lifting attachment		
123	Retaining tool, main bearing shell		
135	Lifting tool, relief valve		
172	Mounting tool, thin bearing shell		
184	Lifting tool, main bearing shell		

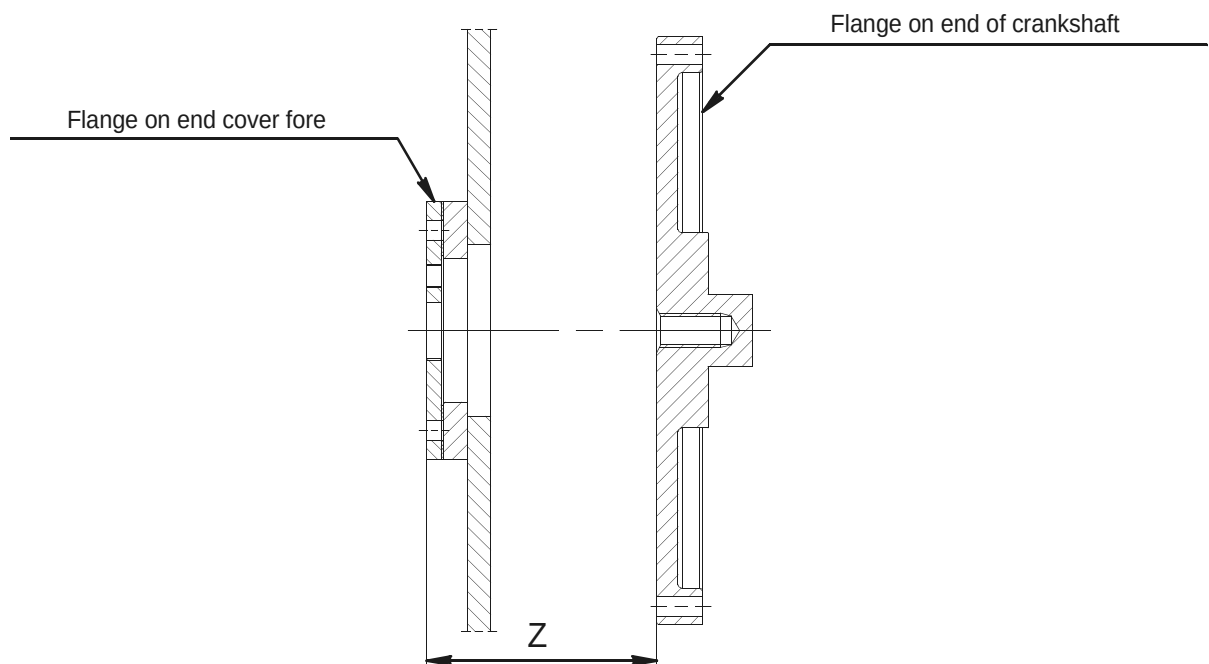
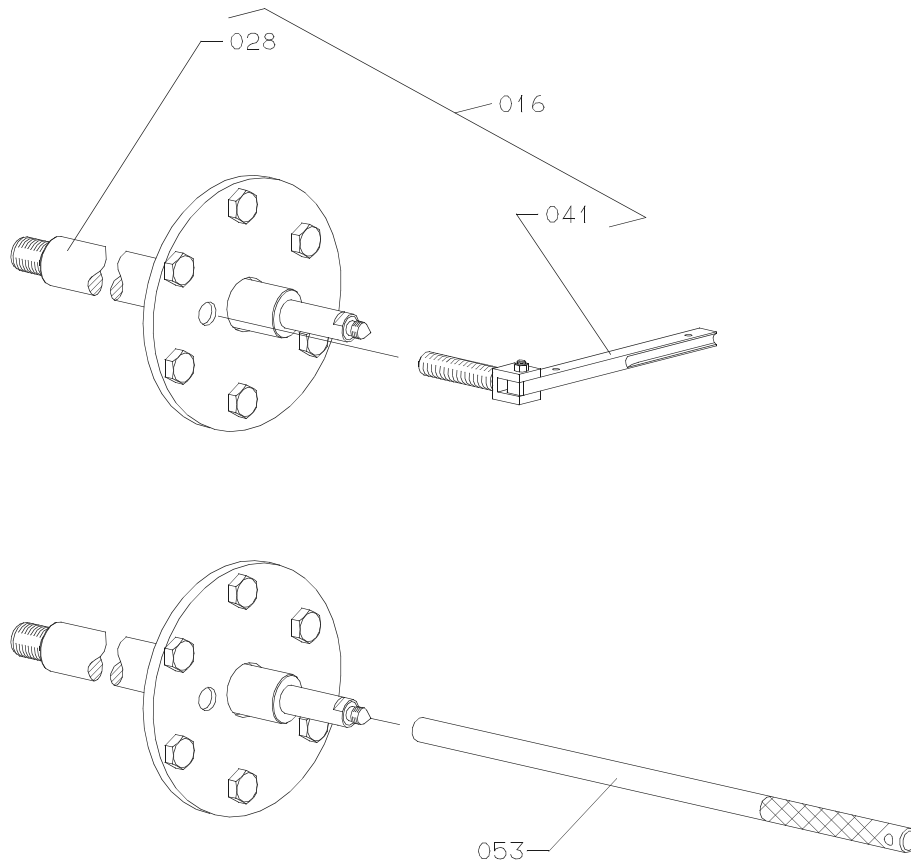


Item No.	Item Description		
021	Jack-hydraulic, complete		
033	Jack-hydraulic, complete		
045	Jack-hydraulic, support		
057	Stud		
069	Nut		
070	Sealing ring with back-up ring		
082	Sealing ring with back-up ring		
094	Sealing ring with back-up ring		
104	Spanner		
130	Ball handle		
141	Lifting tool		
165	Stud setter		
200	Hydraulic tool set, complete		



Item No.	Item Description
015	Support
039	Screw with special head
052	Nut

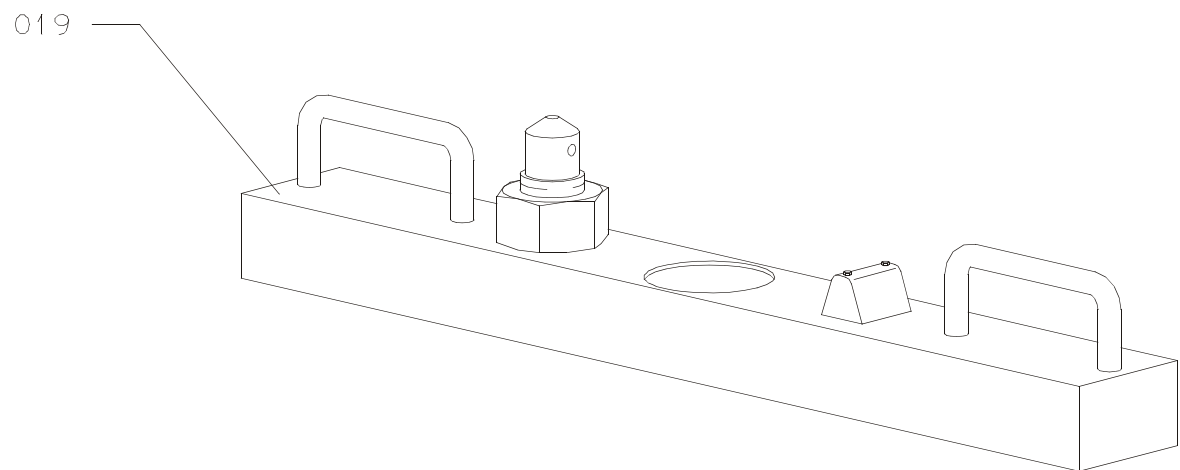
Item No.	Item Description



When ordering item 016 or 028, please always state measurement "Z"

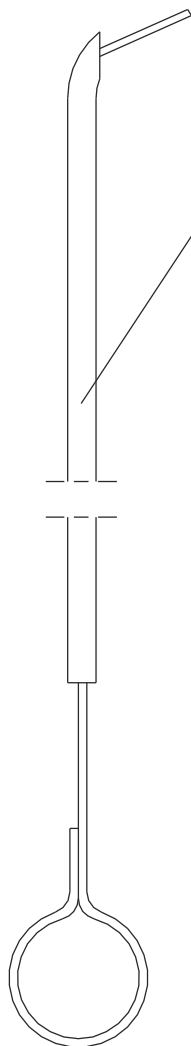
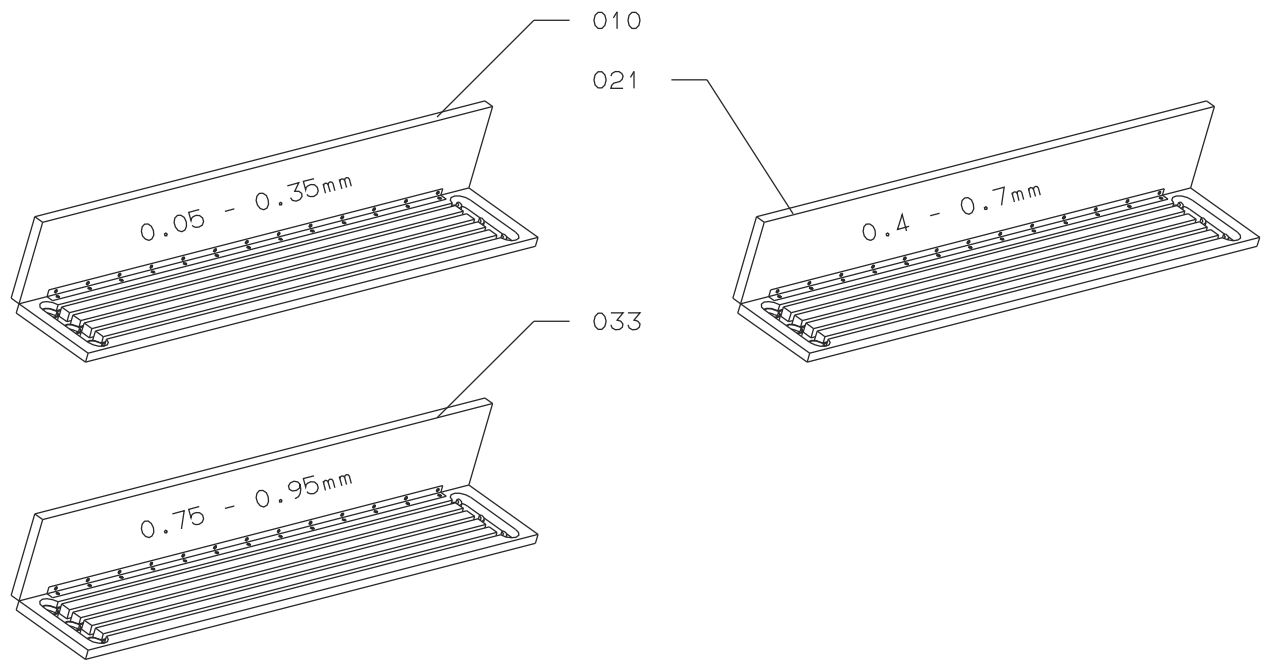
Item No.	Item Description
016	Measuring instruments, axial vibration damper
028	Shaft, measuring instruments axial vibration
041	Measuring instruments, axial vibration damper
053	Handle - sundry types

Item No.	Item Description



Item No.	Item Description
019	Lifting tool for thrust shaft

Item No.	Item Description

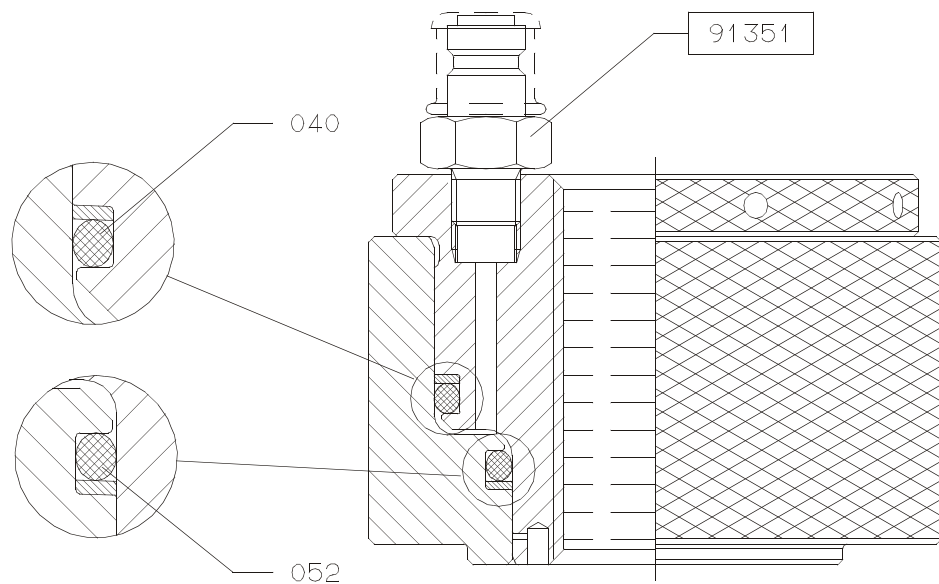
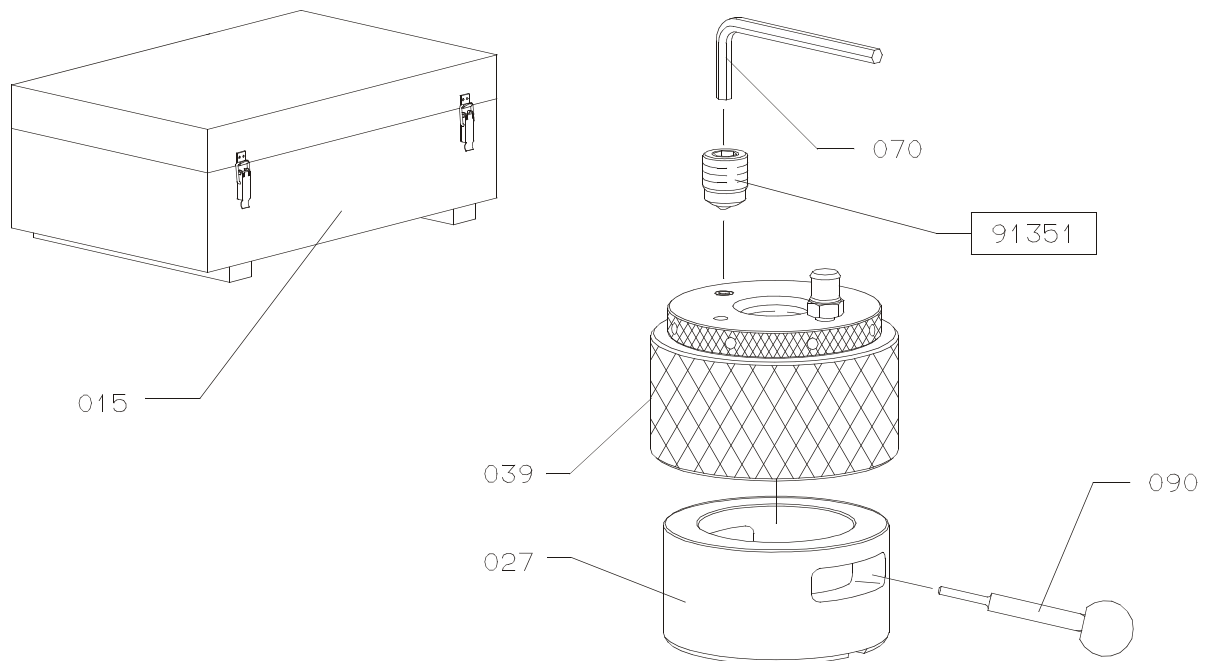


Item	Size (mm)
069	0.05
070	0.10
082	0.15
094	0.20
104	0.25
116	0.30
128	0.35
141	0.40
153	0.45
165	0.50
177	0.55
189	0.60
190	0.65
200	0.70
212	0.75
224	0.80
236	0.85
248	0.90
261	0.95



Item	Size (mm)
332	0.05
344	0.10
356	0.15
368	0.20
381	0.25
393	0.30
403	0.35
415	0.40
427	0.45
439	0.50
440	0.55
452	0.60
464	0.65
476	0.70
488	0.75
511	0.80
523	0.85
535	0.90
547	0.95

Item No.	Item Description		
010	Feeler gauge set		
021	Feeler gauge set		
033	Feeler gauge set		
045	Feeler gauge set		
069	Feeler gauge		
070	Feeler gauge		
082	Feeler gauge		
094	Feeler gauge		
104	Feeler gauge		
116	Feeler gauge		
128	Feeler gauge		
141	Feeler gauge		
153	Feeler gauge		
165	Feeler gauge		
177	Feeler gauge		
189	Feeler gauge		
190	Feeler gauge		
200	Feeler gauge		
212	Feeler gauge		
224	Feeler gauge		
236	Feeler gauge		
248	Feeler gauge		
261	Feeler gauge		
332	Spare tip for feeler gauge		
344	Spare tip for feeler gauge		
356	Spare tip for feeler gauge		
368	Spare tip for feeler gauge		
381	Spare tip for feeler gauge		
393	Spare tip for feeler gauge		
403	Spare tip for feeler gauge		
415	Spare tip for feeler gauge		
427	Spare tip for feeler gauge		
439	Spare tip for feeler gauge		
440	Spare tip for feeler gauge		
452	Spare tip for feeler gauge		
464	Spare tip for feeler gauge		
476	Spare tip for feeler gauge		
488	Spare tip for feeler gauge		
511	Spare tip for feeler gauge		
523	Spare tip for feeler gauge		
535	Spare tip for feeler gauge		
547	Spare tip for feeler gauge		



Item No.	Item Description
015	Wooden box
027	Support
039	Jack-hydraulic, complete
040	O-ring with back-up ring
052	O-ring with back-up ring
076	Key, hexagon socket screw
090	Tommy bar

Item No.	Item Description